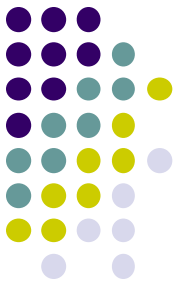




Epigenetics (表观遗传学)

- **Introduction & Definitions**
- **Mechanisms of epigenetics**

Mendelian Genetics

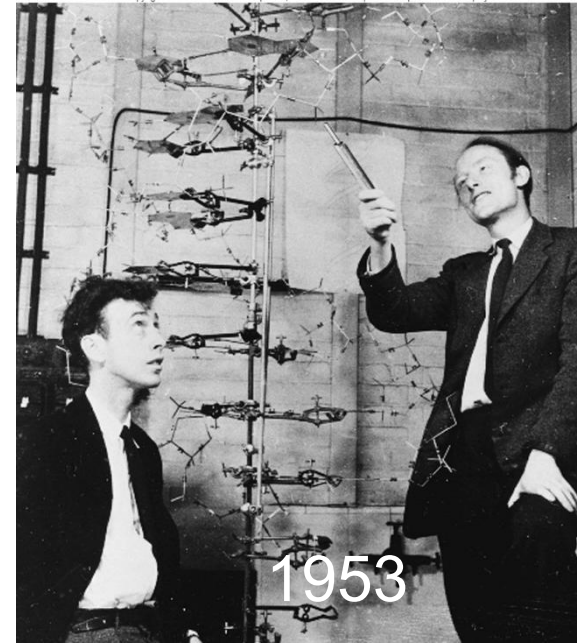


1822-1884
1865



1866-1945
1928

Laws of inheritance
Mendel & Morgan



1953

DNA structure
Watson, Crick, Wilkins, Franklin

**DNA
sequence**



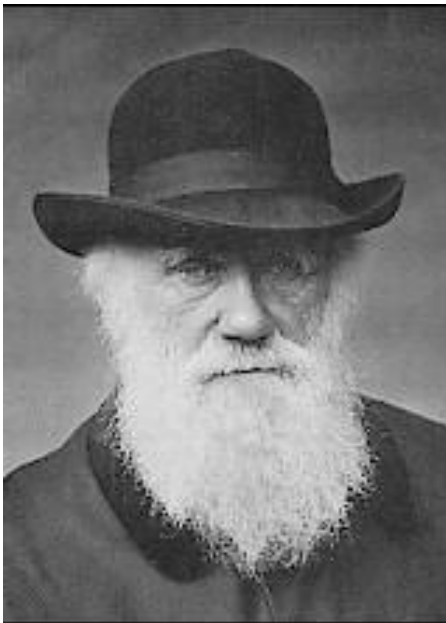
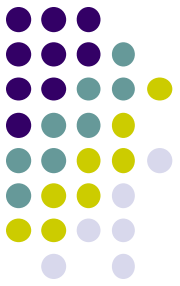
**Gene expression---
protein**



Phenotype

Epigenetics - non-Mendelian genetics

Lamarckian phenomenon

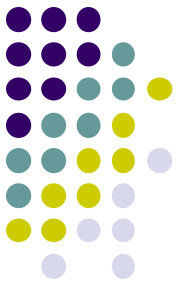


Charles Darwin
(1809-1882)



Jean-Baptiste Lamarck
(1744-1829)

Laws of Lamarckian evolution



- **The law of use and disuse:**

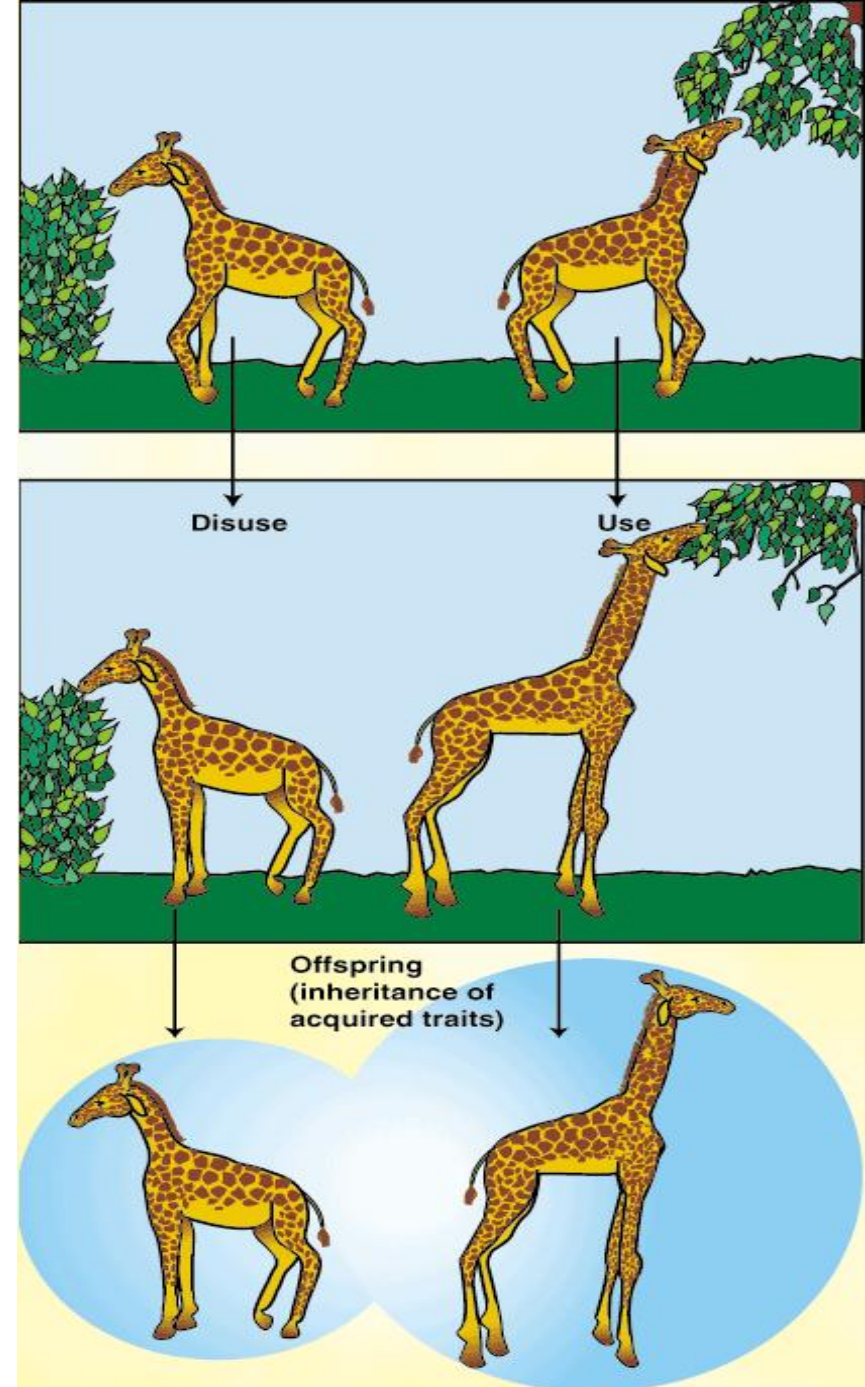
- A more frequent and continuous use of any organ: gradually strengthens, develops and enlarges, and gives it a power proportional to the length of time it has been so used;
- The permanent disuse of any organ: imperceptibly weakens and deteriorates, and progressively diminishes its functional capacity, until it finally disappears.

- **The law of inheritance of acquired characteristics:**

- All the acquisitions or losses wrought by nature on individuals, are preserved by reproduction to the new individuals which arise
- The acquired modifications are common to both sexes, or at least to the individuals which produce the young.

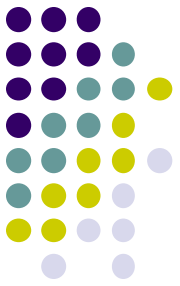
Evidence for Lamarckism

- Giraffes stretching their necks to reach leaves high in trees (especially Acacias), strengthen and gradually lengthen their necks.
- These giraffes have offspring with slightly longer necks
- The phenomena is also known as "soft inheritance"



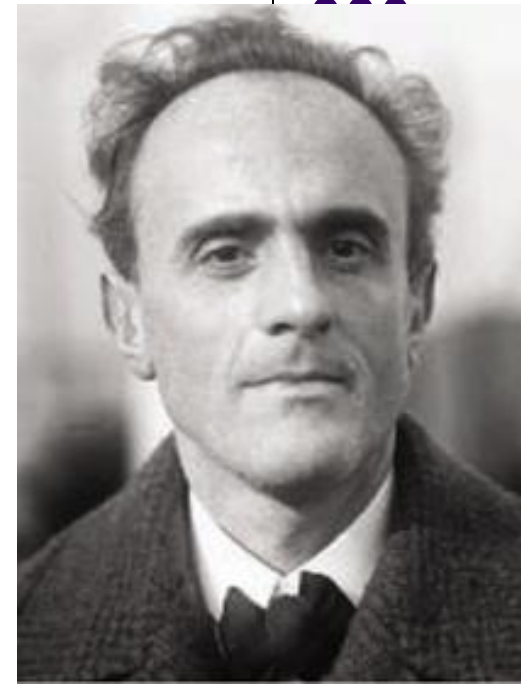
The evolution of Giraffe necks (Jean-Baptiste Lamarck)

Traditional examples considered as Lamarckian inheritance



- A blacksmith, through his work, strengthens the muscles in his arms.
- His sons will have similar muscular development when they mature.

Story of Midwife toad

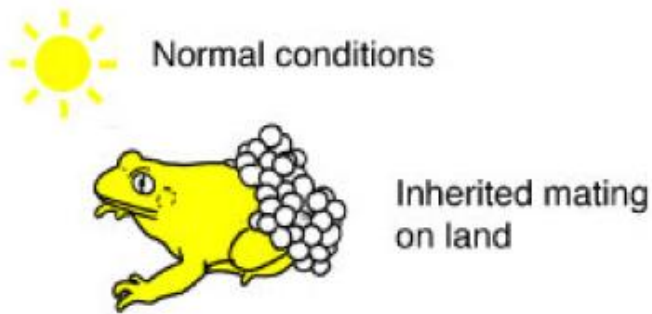


Paul Kammerer
(1880-1926)

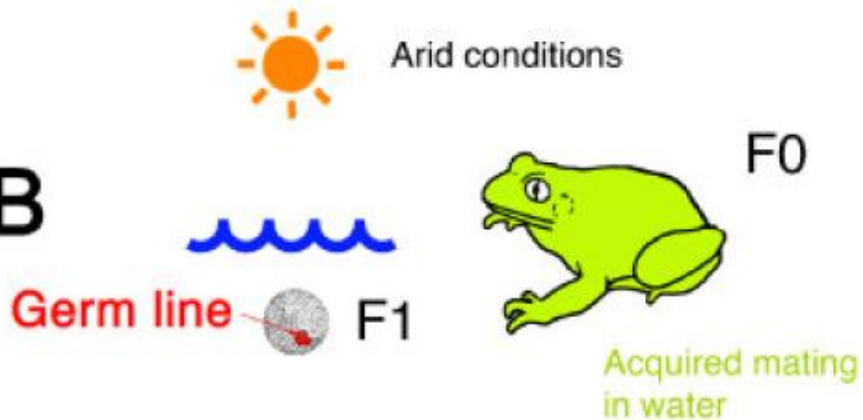
- Most frogs and toads mate in water and lay their eggs in aquatic environments.
- Midwife toads mate on dry land, and males subsequently wrap the strings of eggs around their legs, carrying them around until the embryos are ready to emerge as tadpoles.



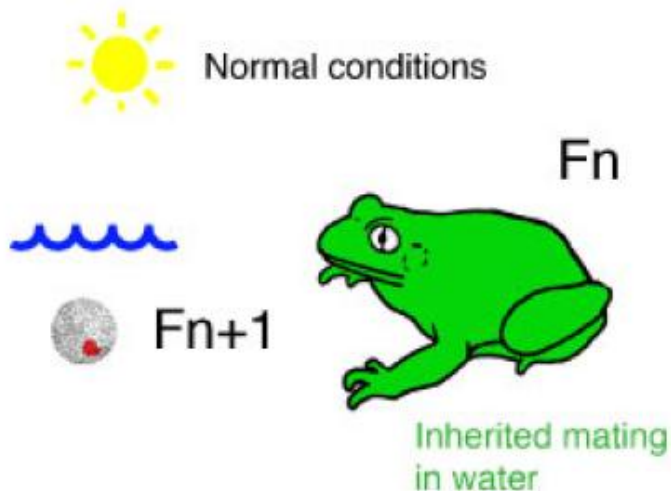
A



B



C



- Kammerer confined the toads to a dry, overheated environment, driving them to mate and lay their eggs in water.
- Most of the eggs died, but the 3% to 5% of offspring that survived had lost the terrestrial habits of their parents.
- Even in cool, moist environments, they opted to mate and deposit their eggs in water, a preference that Kammerer said persisted for at least six generations, the amount of time Kammerer studied them.

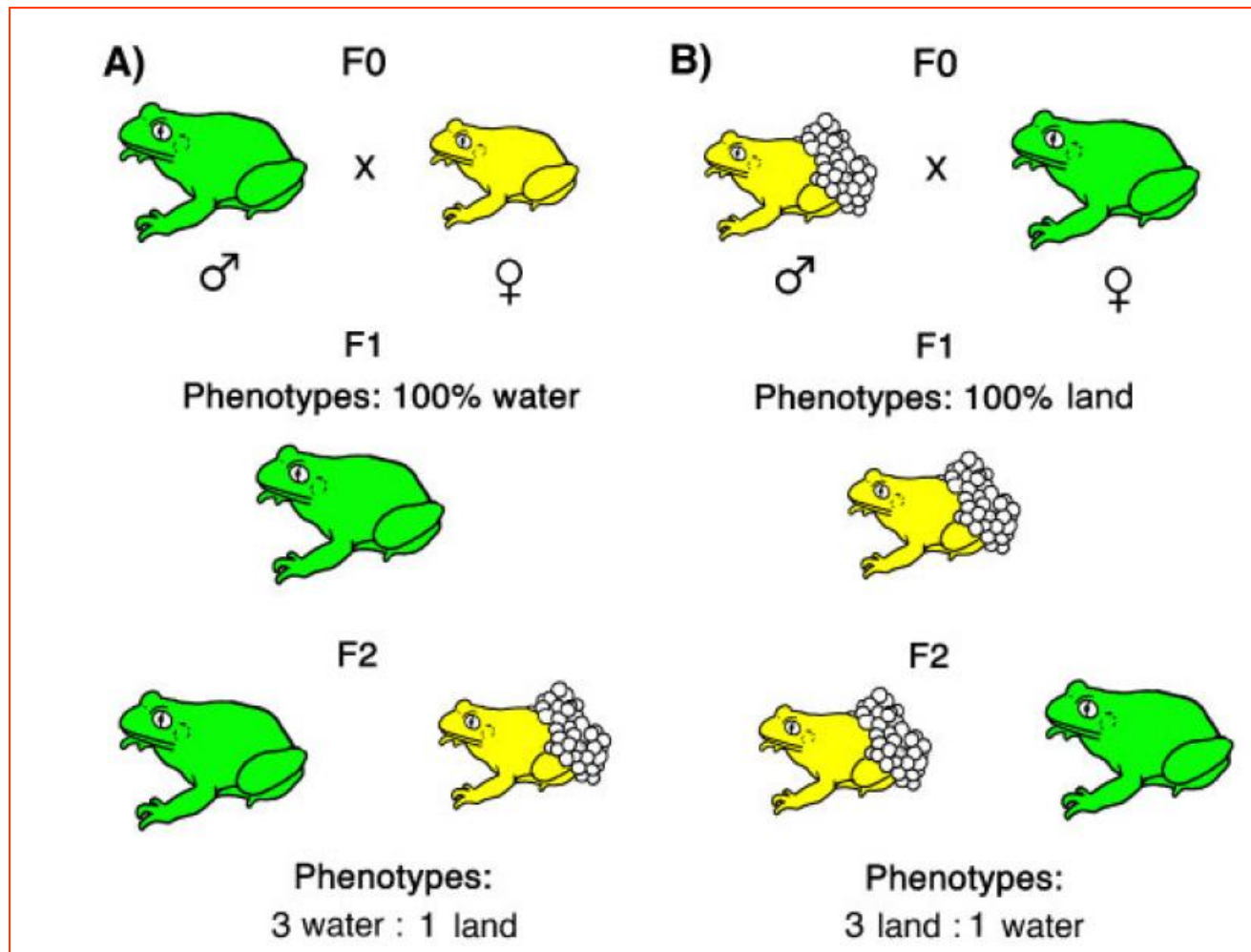
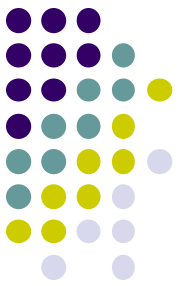


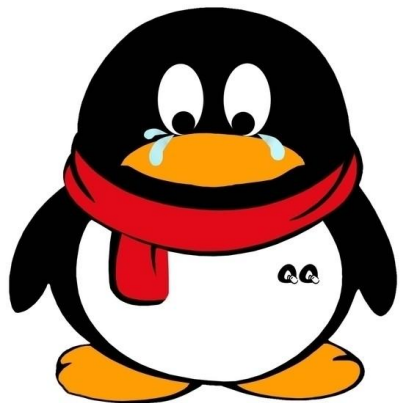
Fig. 2. Parent-of-origin effects in Kammerer's hybrid crosses.

Victim of his time?



- Kammerer reported that by generation F4, “water” males were developing noticeable nuptial pads on its foot - an acquired characteristic brought about by adaptation to environment.
- **Controversy:** specimen distributed.
- Claims arose that the result of the experiment had been falsified. The most notable of these claims was made by Dr. Noble in the scientific journal Nature in 1926.

Dr. Noble reported that the black pad actually had a far more mundane explanation: it had simply been injected there with Indian ink.



Six weeks later, **Kammerer committed suicide.**

Science 4 September 2009:

Vol. 325. no. 5945, pp. 1194 - 1195

DOI: 10.1126/science.325_1194



The Case of the Midwife Toad: Fraud or Epigenetics?

Elizabeth Pennisi

Paul Kammerer has been called the perpetrator of one of the most celebrated scientific frauds of the early 20th century. He has also been defended as the victim of forgery by a lab assistant; some even say he was framed by political or scientific opponents. Now, a new analysis published this week suggests his infamous experiments may have been the first demonstrations of a recently recognized phenomenon: epigenetics.

Did Paul Kammerer Discover Epigenetic Inheritance? A Modern Look at the Controversial Midwife Toad Experiments

ALEXANDER O. VARGAS*

Laboratory of Ontogeny and Phylogeny, Department of Biology, Faculty of Science, University of Chile, Las Palmeras, Ñuñoa, Casilla, Santiago, Chile

- A re-examination of Kammerer's descriptions of hybrid crosses of treated and nontreated toads reveals **parent-of-origin effects** like those documented in epigenetic inheritance.
- Modification of the extracellular matrix of the egg provides a plausible cause for altered **gene methylation patterns**. Traits such as altered egg and adult body size in Kammerer's "treated" toads are inherited epigenetically in other tetrapods.

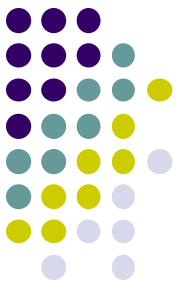
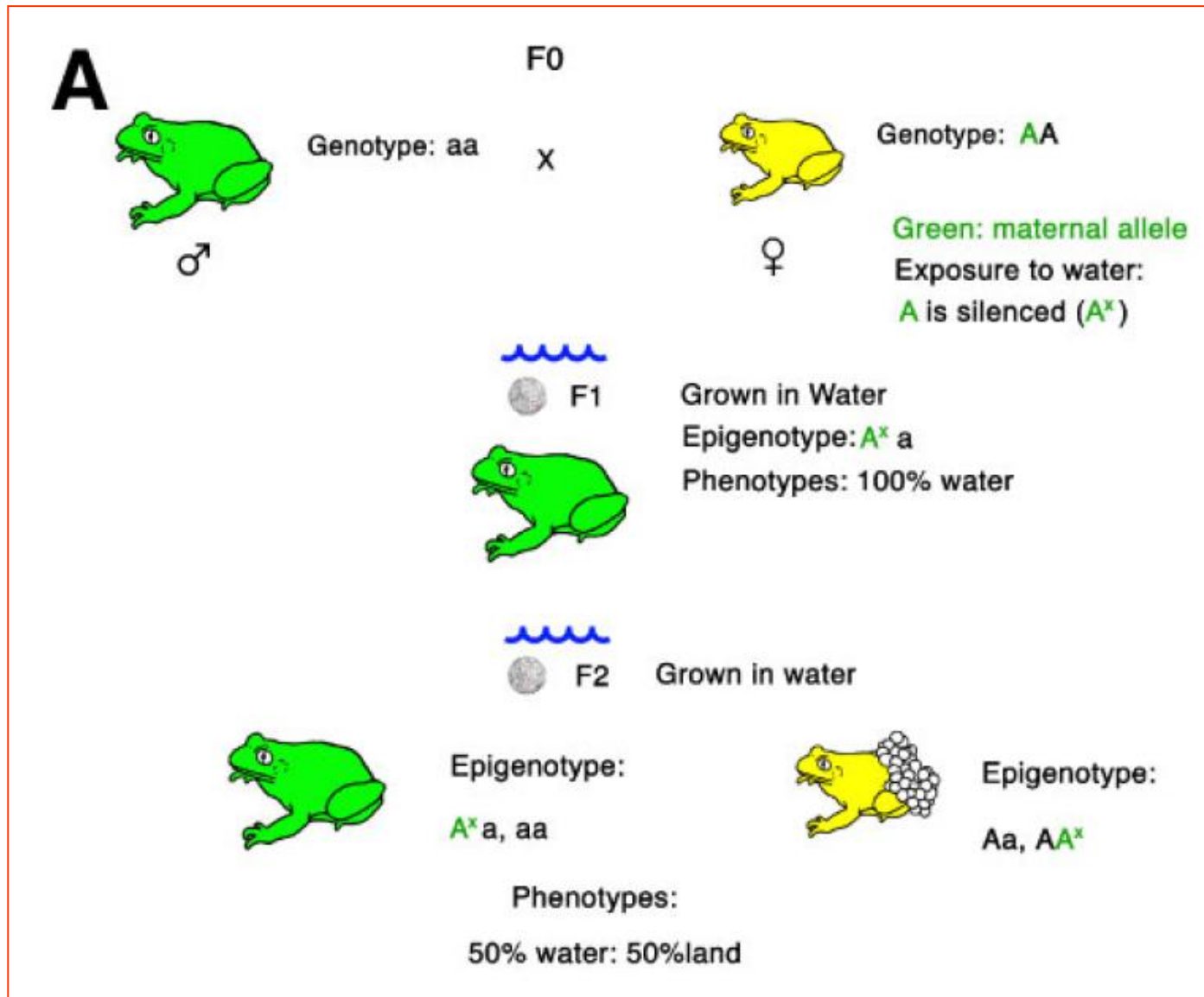


Fig. 3. A preliminary epigenetic model for Paul Kammerer's hybrid cross using a male “water” toad.

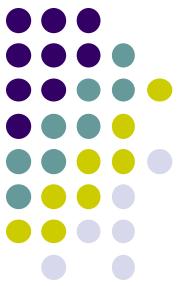
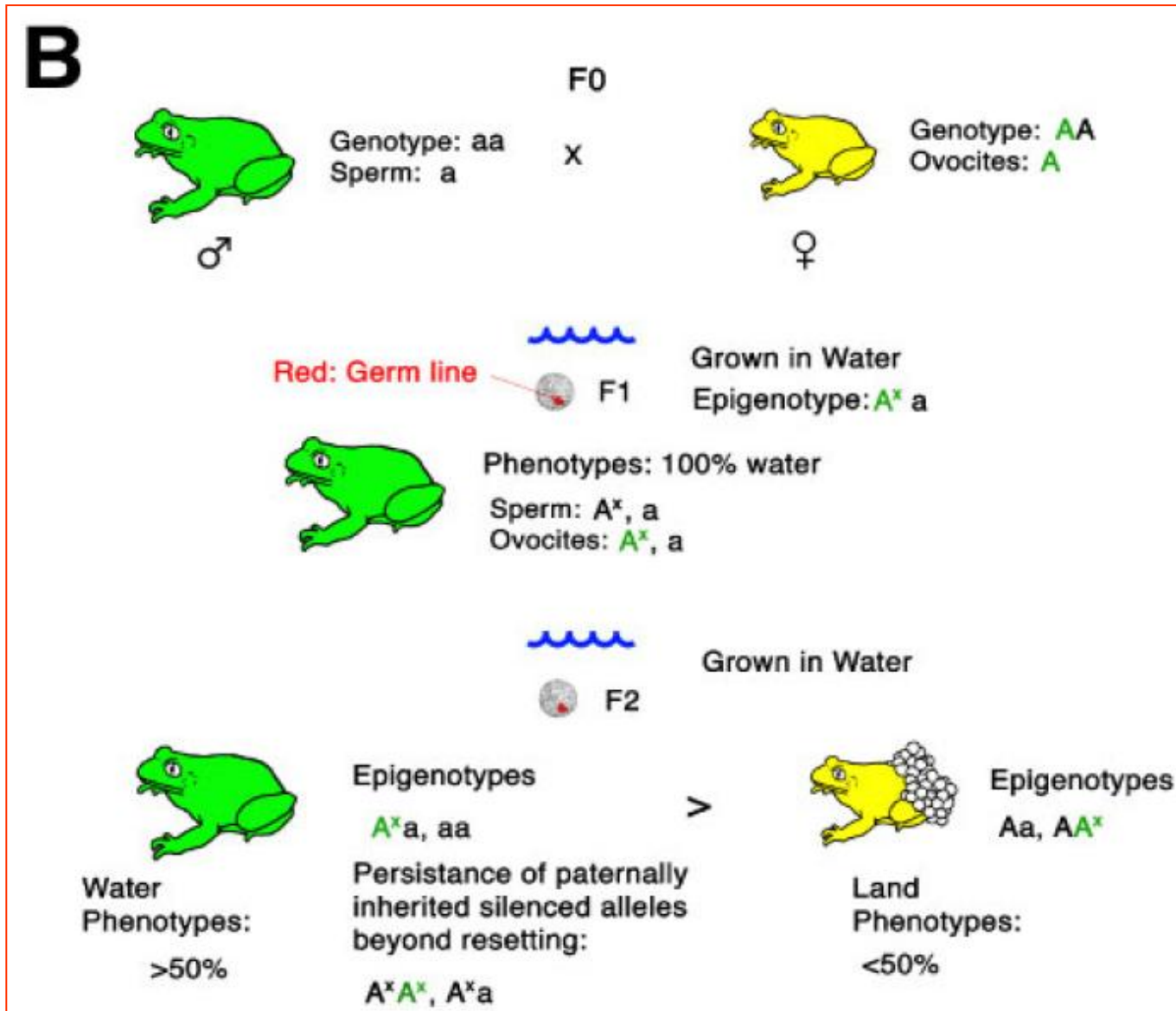


Fig. 3. A preliminary epigenetic model for Paul Kammerer's hybrid cross using a male "water" toad.

Original wild stock:

Aa and AA genotypes

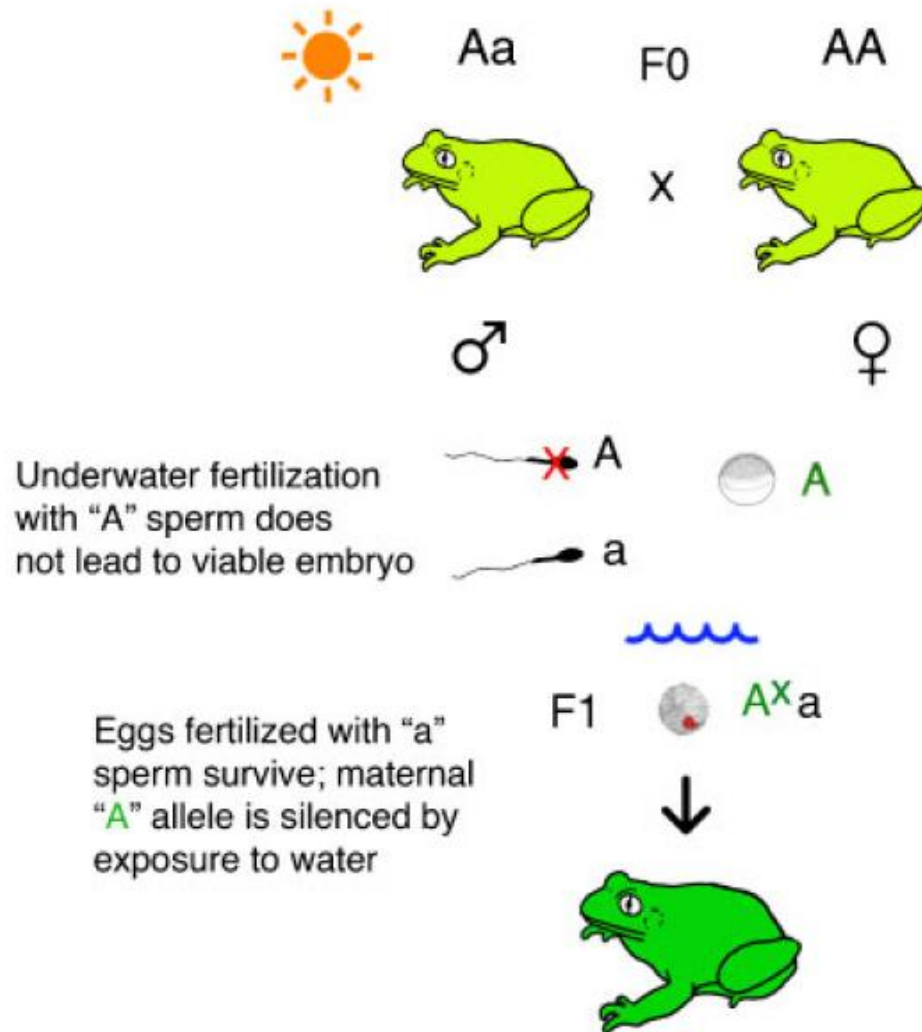


Fig. 4. A hypothesis on the difference of fertilization as exposed to water or air in Kammerer's basic experiment (Fig. 1). When exposed to water, only eggs fertilized with sperm carrying an inactive "a" allele can develop to term. Only a few individuals in Kammerer's initial population of land toads would carry the rare "a" allele. This may explain the low rate of egg survival described by Kammerer in F1. The few surviving F1 toads would carry an "a" paternally inherited allele and a maternally inherited "A" allele that is silenced upon exposure to water, developing the "water" phenotype.

Types of inheritance



- **Hard inheritance:** essentially Mendelian; hereditary material remains constant between generations (except random mutations)
- **Soft inheritance (软式遗传):** the generation of a new phenotype is less rigidly determined and shows a more rapid response to environment
 - Soft inheritance is adaptive in the sense used by evolutionary biologists
 - The ability of epigenetic mechanisms to perpetuate gene expression patterns relatively stably, and to retain the capacity to react to environmental cues, makes them ideal for facilitating soft inheritance.
 - The largest barrier to such a system is the resetting of epigenetic marks between generations.
- **Parental effects:** effects on the phenotype of offspring that are not determined by the offspring's own genotype but by the genotype or environmental experience of its parents

Definitions

The term 'Epigenetics' was introduced by Conrad H. Waddington in 1942 to describe “The interactions of genes with their environment that bring the phenotype into being”.

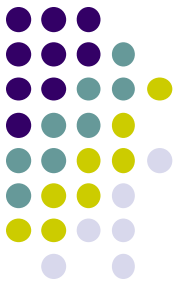
The current concept of

Epigenetics:
Changes in phenotype that are inheritable
but do not involve DNA mutation.



***Conrad H. Waddington
(1905-1975)***

Soft genetics and Lamarckian evolution



- Lamarck: both environment and behavior direct organic change in an organism's form and guide adaptation through the generations.
- A major problem with Lamarckian evolution is the separation of **germline and soma**.

Science *7 April 2000:Vol. 288. p. 38*



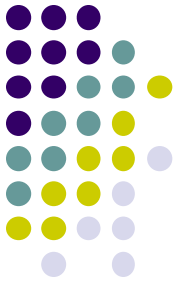
Was Lamarck Just a Little Bit Right?

Michael Balter

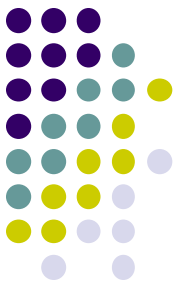
Although Jean-Baptiste Lamarck is remembered mostly for the discredited theory that acquired traits can be passed down to offspring, new findings in the field of epigenetics, the study of changes in gene expression that are not linked to alterations in DNA sequences, are returning his name to the scientific literature. Although these new findings do not support Lamarck's overall concept, they raise the possibility that "epimutations," as they are called, could play a role in evolution.

Lamarck was a true pioneer of evolutionary theory!

Epigenetics (表观遗传学)



- **Introduction & Definitions**
- **Mechanisms of epigenetics**



Stories of Agouti mice (豚鼠)

Dr. Jirtle and Waterland , Duke University

These Two Mice are Genetically Identical and the Same Age



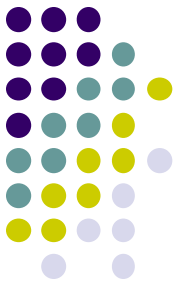
While pregnant, both of their mothers were fed Bisphenol A (BPA) but **DIFFERENT DIETS**:

The mother of this mouse received a **normal mouse diet**

The mother of this mouse received a diet **supplemented** with choline, folic acid, betaine and vitamin B12



Agouti mice

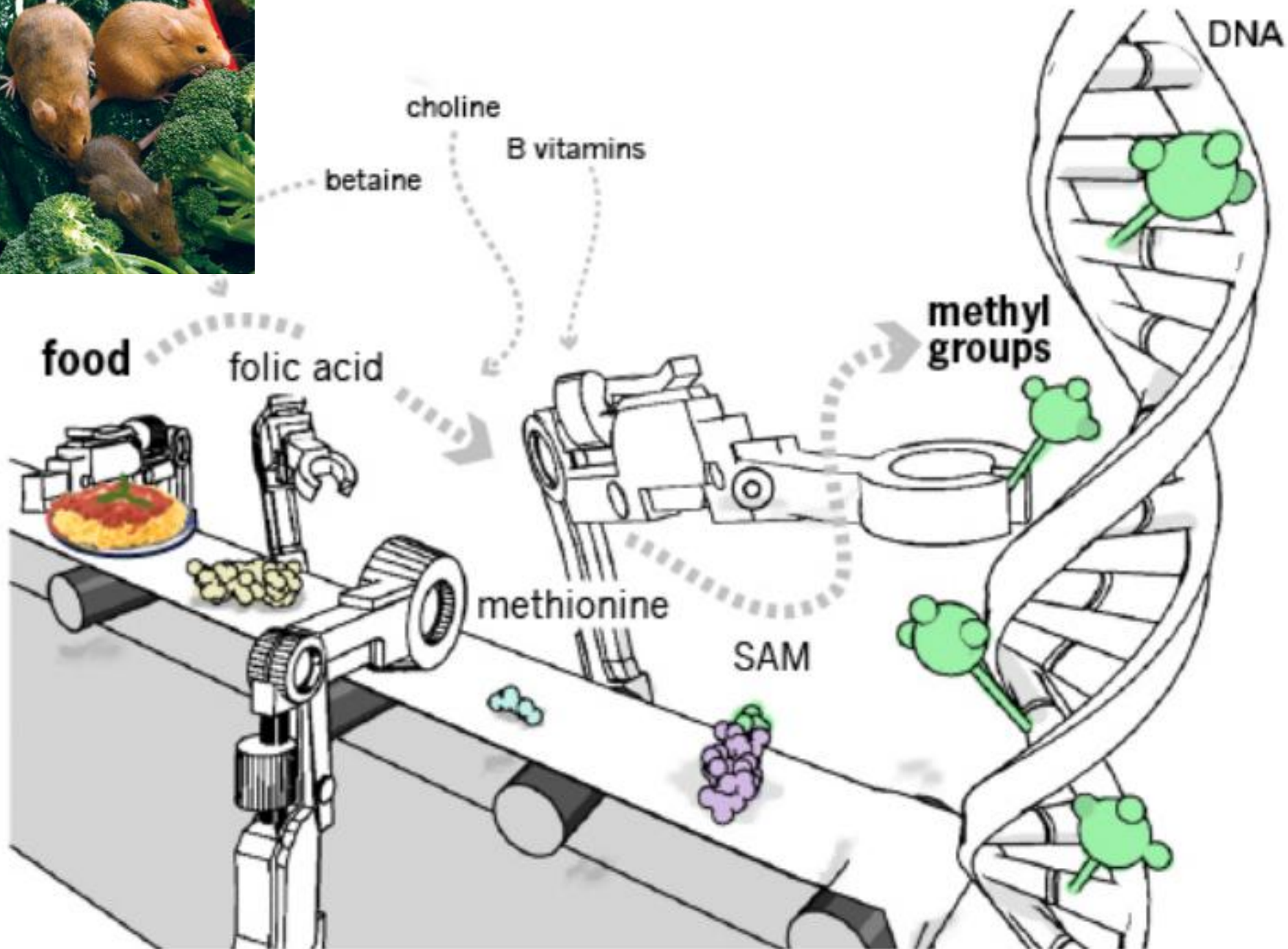


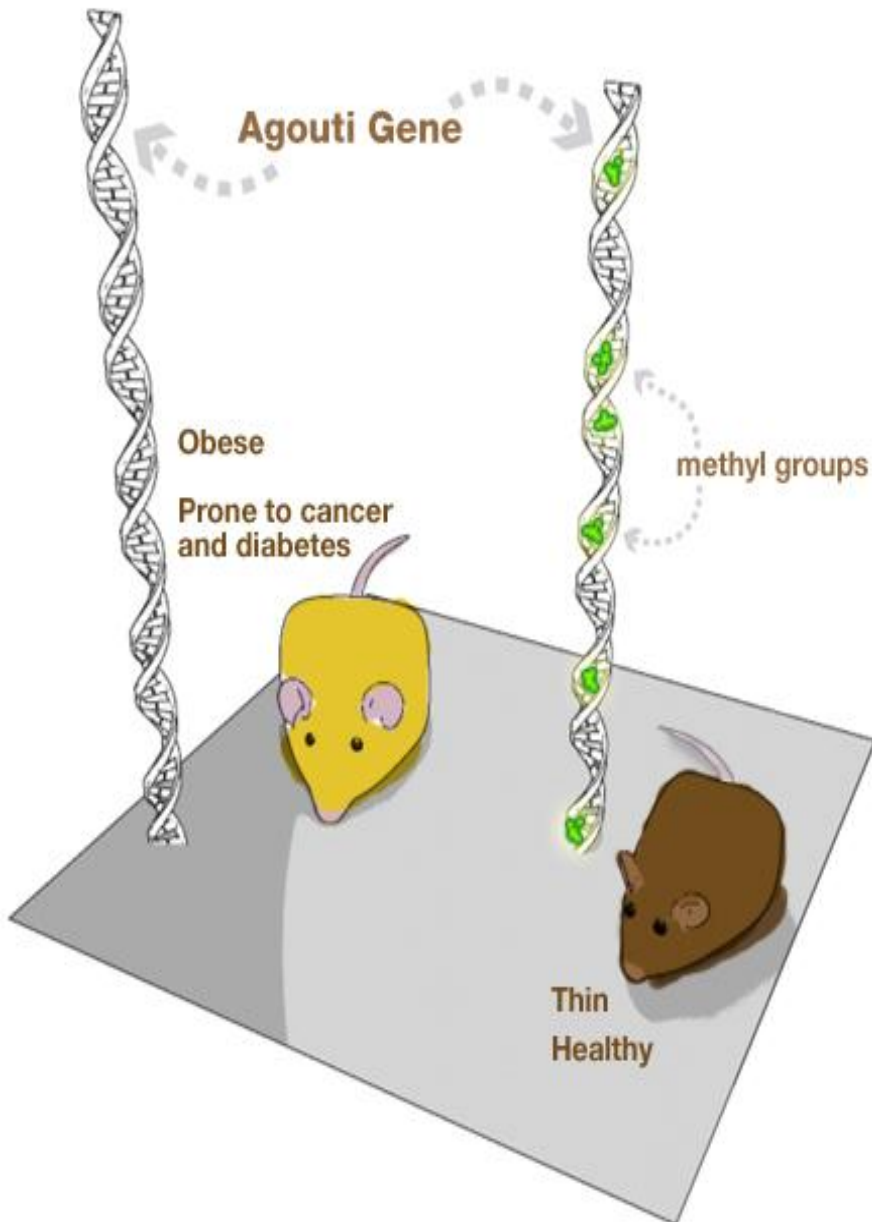
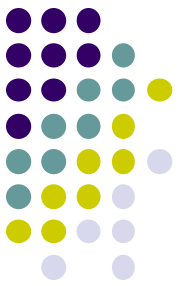
- **DNA Is Not Destiny :**
 - **Fat yellow mice** known to scientists as **agouti mice**, so called because they carry a particular gene—the agouti gene—that in addition to making the rodents ravenous and yellow renders them prone to cancer and diabetes.
 - Typically, when agouti mice breed, most of the offspring are identical to the parents: just as yellow, fat as pincushions, and susceptible to life-shortening disease.
 - The parent mice in Jirtle and Waterland's experiment, however, produced a majority of offspring that looked altogether different. These young mice were slender and mousy brown. Moreover, they did not display their parents' susceptibility to cancer and diabetes and lived to a spry old age. The effects of the agouti gene had been virtually erased.

What make the transformation?



- Without altering a single letter of the mouse's DNA.
- Only changed the moms' diet
 - Fed a test group of mother mice a diet rich in methyl donors.
 - These molecules are common in the environment and are found in many foods, including onions, garlic, beets, and in the food supplements often given to pregnant women.
 - After being consumed by the mothers, the methyl donors worked their way into the developing embryos' chromosomes and onto the critical agouti gene.
 - The mothers passed along the agouti gene to their children intact, but thanks to their methyl-rich pregnancy diet, they had added to the gene a chemical switch that dimmed the gene's deleterious effects.





These Two Mice are Genetically Identical and the Same Age

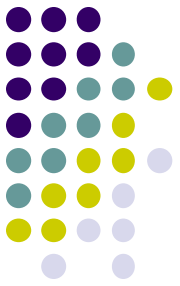


While pregnant, both of their mothers were fed Bisphenol A (BPA) but **DIFFERENT DIETS**:

The mother of this mouse received a **normal mouse diet**

The mother of this mouse received a diet **supplemented** with choline, folic acid, betaine and vitamin B12

Diets changes the genetics of the mother mice



- Yellow fur and brown fur mouse are genetics identical, but the ectopic agouti expression is different, depending on the methylation of agouti gene
- Maternal diet during pregnancy affects the methylation of the gene.
 - Folic acid or other methyl-rich diet, resulted in brown health offspring
 - Exposure to Bisphenol A turns on the gene at wrong time or in wrong tissue

A Bee's Royal Diet



- **Royal jelly:** complex, protein-rich substance secreted from glands on the heads of worker bees.
- **Workers and queens are genetically identical**
 - With the royal jelly diet, the queen will develop functional ovaries and a larger abdomen for egg laying, while worker bees remain sterile.
 - Royal jelly silences a key gene (Dnmt3), which codes for an enzyme involved in genome-wide gene silencing.



- When Dnmt3 is active, the queen genes are epigenetically silenced and the larvae develop into the default “worker” variety.
- When royal jelly turns Dnmt3 off, certain genes jump into action that turn the lucky larvae into queens.

Kucharski R., Maleszka J., Foret S., Maleszka R. Nutritional Control of Reproductive Status in Honeybees via DNA Methylation (2008). Science, 319: 1827 - 1830.

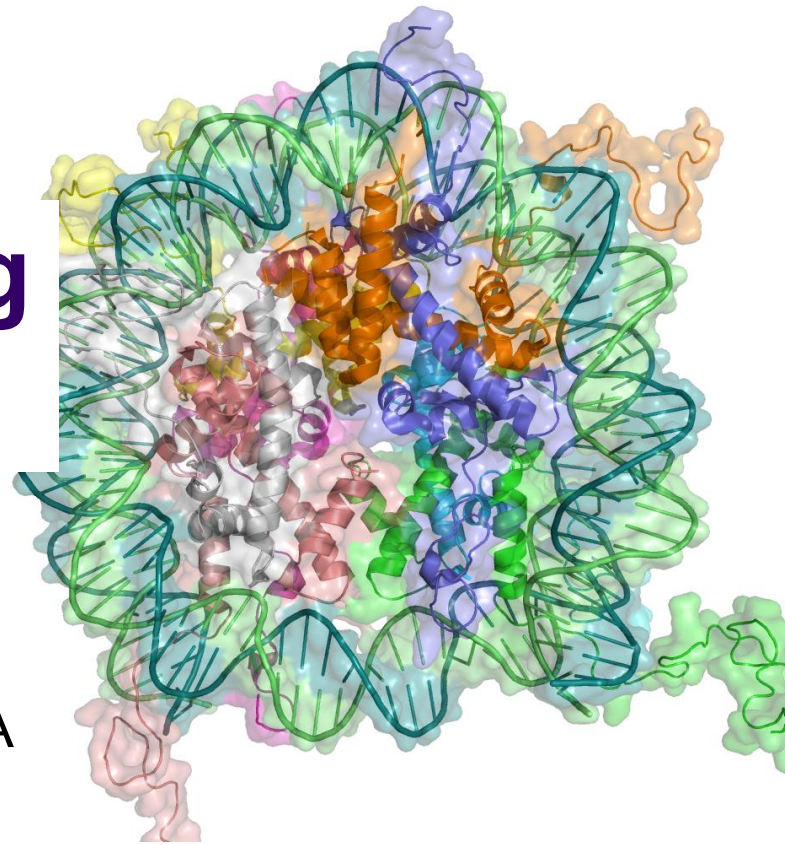


Mechanism of epigenetics

1. DNA methylation
2. Histone modification and chromatin remodeling
3. Small RNAs
4. Prions
5. Others

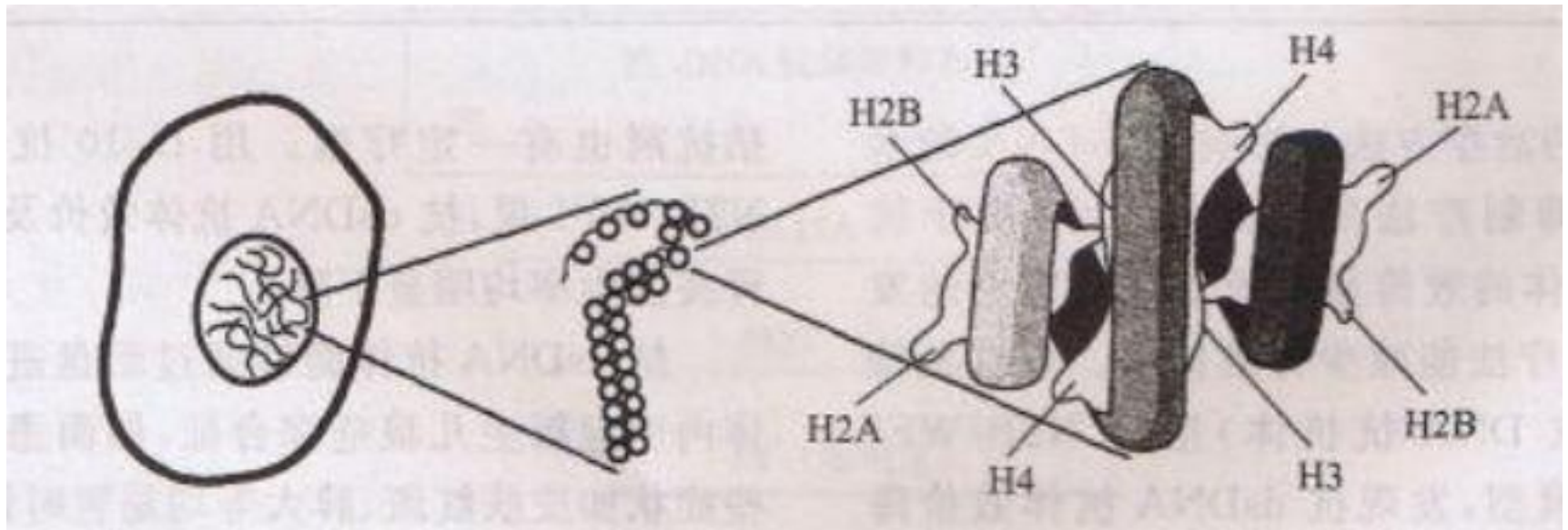
Chromatin remodeling

- **Chromatin**= DNA + the histone proteins
- Histone proteins are little spheres that DNA wraps around.
- If the way that DNA is wrapped around the histones changes, gene expression can change as well.





组蛋白有五种H1、H2A、H2B、H3和H4，在真核细胞中，组蛋白与DNA紧密结合形成核小体。核小体中心由(H3—H4)₂四聚体组成，H2A—H2B二聚体位于其两侧。核小体被大约146个碱基对的DNA缠绕，多个核小体串联在一起，中间靠H1(有20~60个碱基对DNA缠绕)连接，这种结构再进一步弯曲、折叠即形成染色质。用微球菌核酸酶和3M脲素消化核小体，分解出3个亚基：H1与60—70个碱基对DNA复合体；H2A—H2B与50—60个碱基对DNA复合体以及(H3—H4)₂与70~80个碱基对DNA复合物。在合适的条件中，这些组蛋白—DNA复合体处于可溶性稳定状态。



Mechanisms of chromatin remodeling



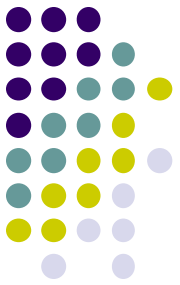
- **Modification of histone proteins:**

Histone proteins are made up of long chains of amino acids. If the amino acids that are in the chain are changed, the shape of the histone sphere might be modified. By altering the shape of the histones around it, these modified histones would ensure that a differentiated cell would stay differentiated, and not convert back into being a stem cell.

- **Methylation of DNA, at CpG sites:**

to convert cytosine to 5-methylcytosine. Cytosine is the nucleotide that our cells can "read." Our cells cannot "read" methylcytosine. If DNA is conceived as an instruction manual again, changing cytosine to methylcytosine, the cell can no longer "read" the gene. Thus, the gene is turned off.

Histone code

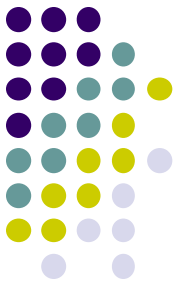


- **Histone code** (组蛋白密码): 组蛋白在翻译后的修饰中会发生改变, 提供一种识别的标志, 为其他蛋白和DNA的结合产生不同效应 (协同或拮抗)。是一种动态转录调控成分。

Modifications of the Histone tails act as marks that can be read by other proteins to control the expression or replication of chromosomal regions. The coding in the histones may be heritable.

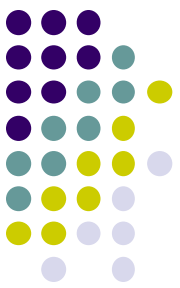
Generally, histone acetylation is associated with transcriptionally active genes

Deacetylation is associated with inactive genes (= gene silencing)



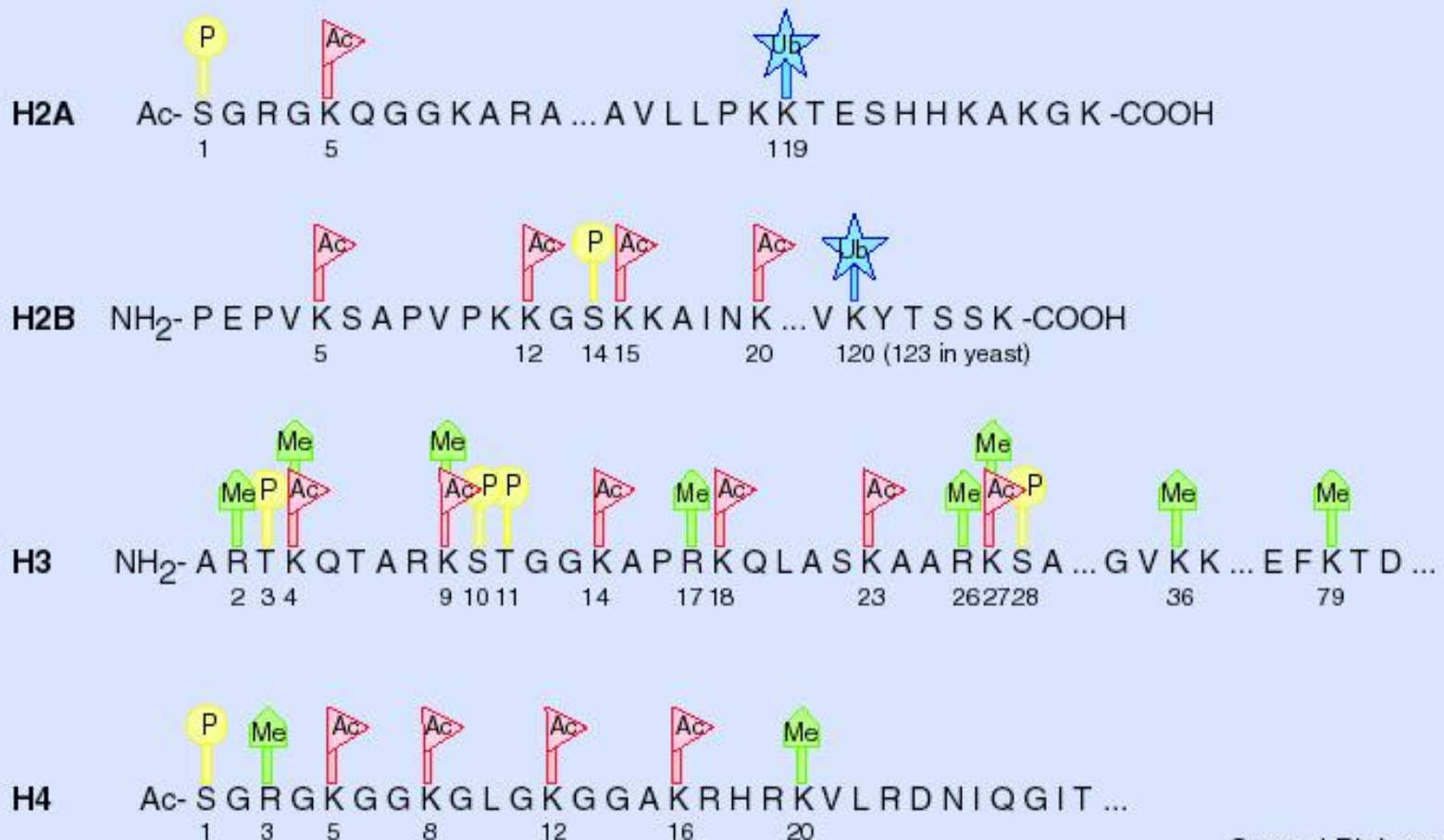
Histone modifications

- Throughout the entire sequence, particularly in the unstructured termini of histones - “histone tails”
- Including:
 - Acetylation: associated with "active" transcription
acetylation of the K14 and K9 lysines of the tail of histone H3 by histone acetyltransferase enzymes (HATs) is generally correlated with transcriptional competence
 - Methylation:
 - Ubiquitylation
 - Phosphorylation
 - Sumoylation

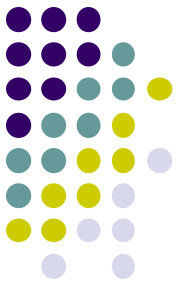


Histone Modifications

- Different combinations of histone modifications, especially located near or within a gene's promoter, may be VERY SPECIFIC to the transcriptional state of the gene



Enzymes providing histone modifications



Acetylation: HATs - CBP, p300, GCN5, ATF2, Tip 60...

Deacetylation: HDACs- class I and II

Methylation:

- ***Lysine*** : SET-domain HMTase and non-SET domain HMTase (Dot1)

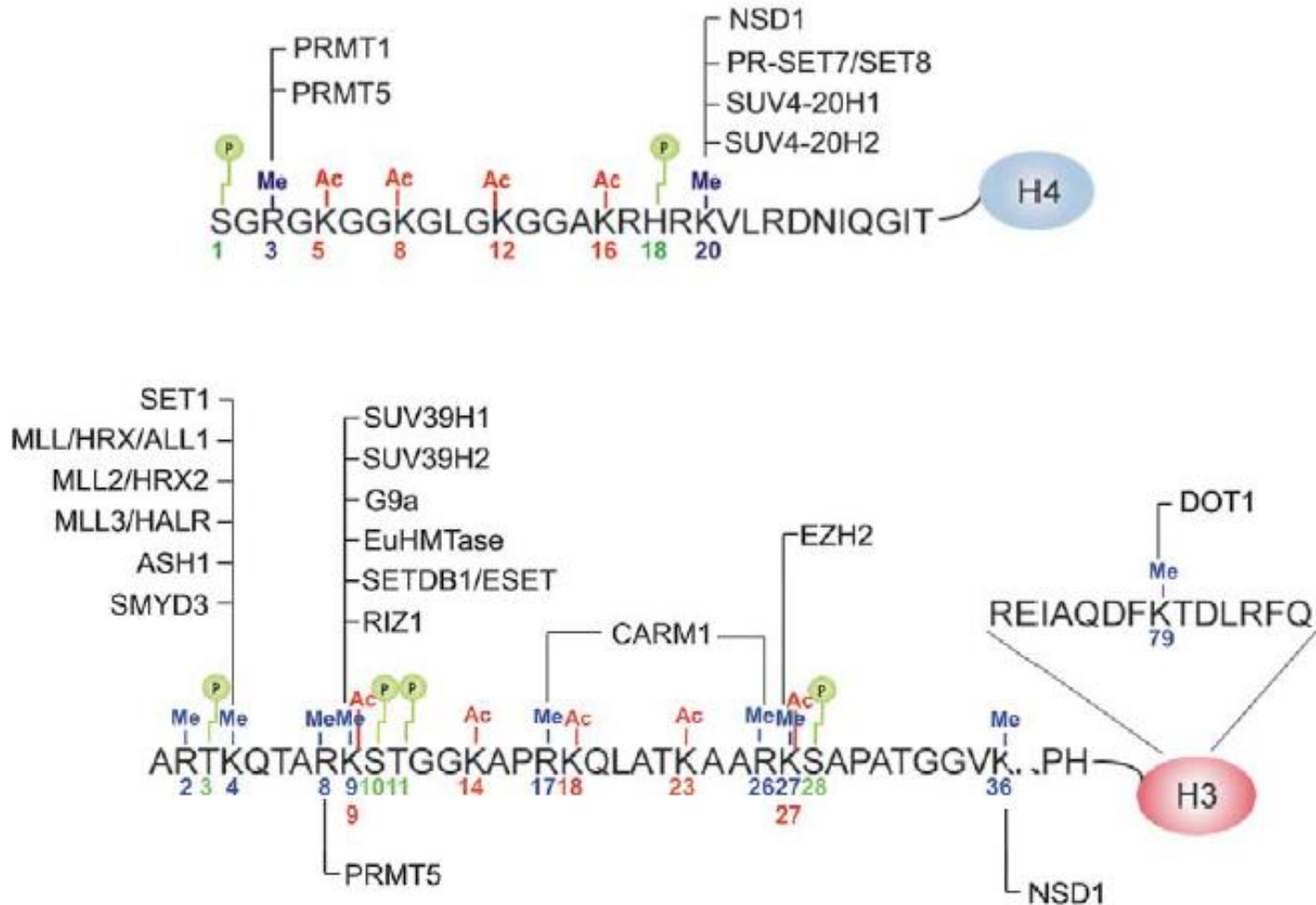
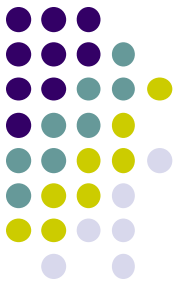
- ***Arginine:*** PRMT family, CARM1

Demethylation: LSD1 (demethylate mono- and di-methylated lysines)

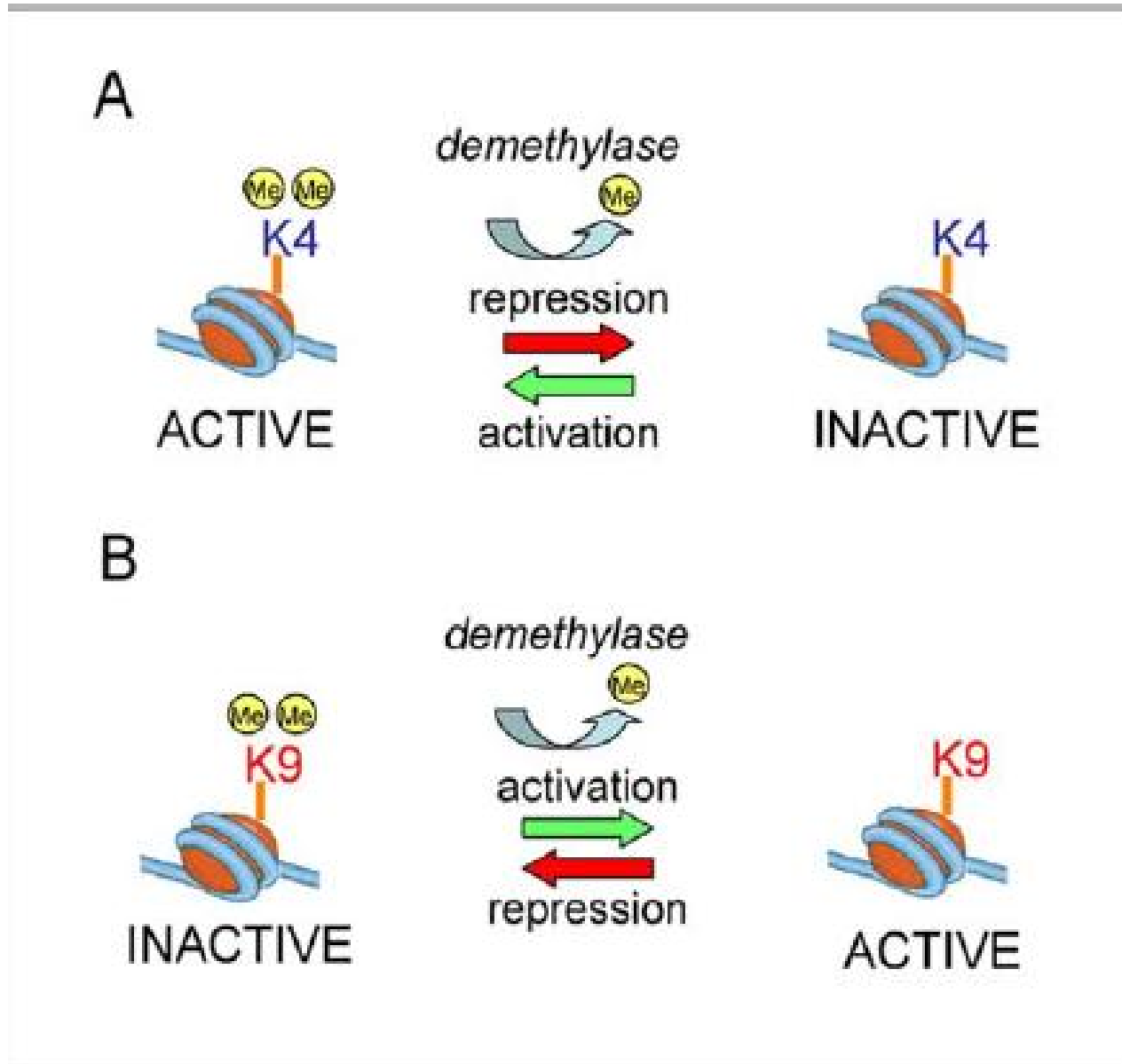
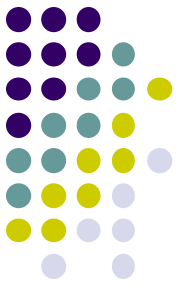
Ubiquitination: ubiquitin conjugase Rad6/ligase Bre1 for H2B

De-Ubiquitination: SAGA-associated Ubp10

Enzymes providing histone methylation



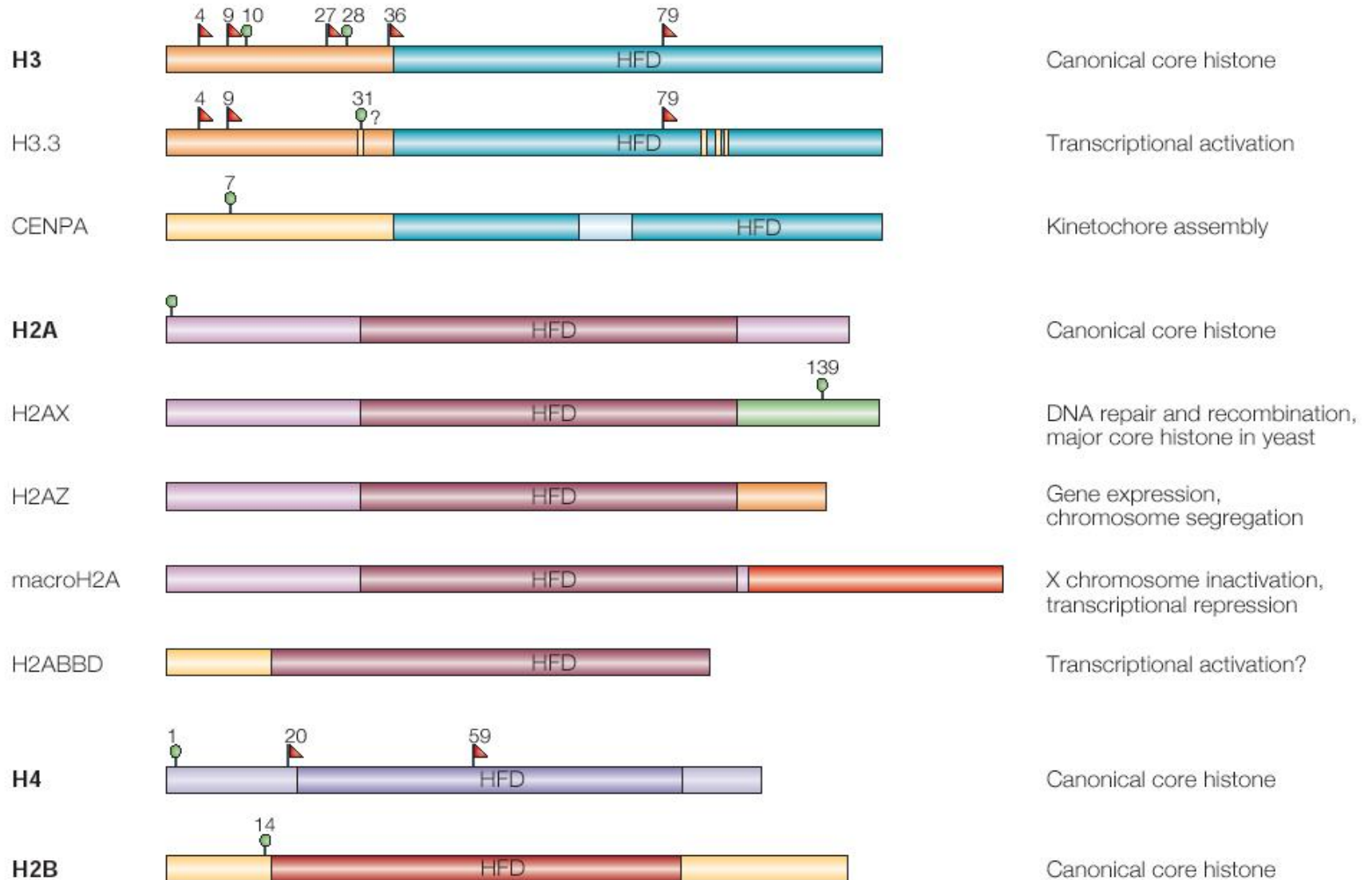
Histone demethylase LSD-1



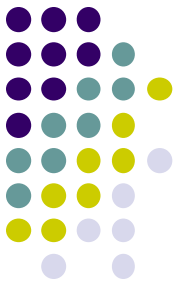
Shi et al., (2004).

Metzger et al. (2005)

Histone variants



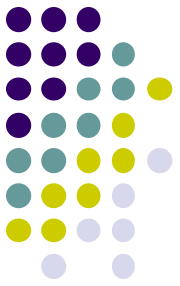
Connection between histone variants and modifications



Example: The modifications shown by ChIP to be enriched in active chromatin, including acetylation of K9, K14, K18 and K23 and methylation of K4 and K79, were found to be enriched two to fivefold in H3.3 relative to H3.

Variant histones are assembled by specific complexes

Example: The SWR1(Snf2/Snf2-related ATPase) complex appears to destabilize the nucleosome, exchanging H2A.Z–H2B for H2A–H2B.



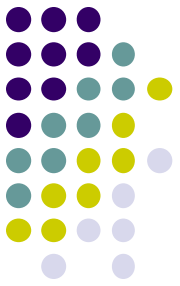
Histone code hypothesis

Something is writing this code....

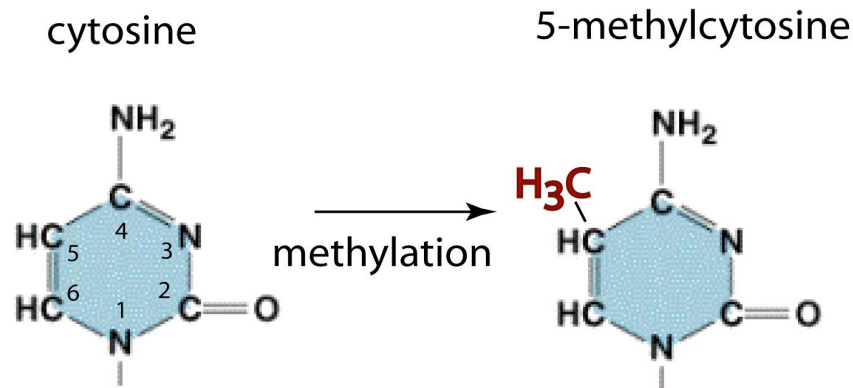
and

Something is reading this code...

DNA Methylation and CpGs sites

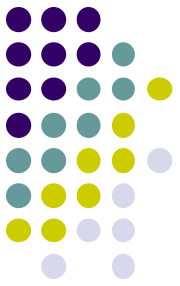


- **DNA methylation**: a type of chemical modification of DNA that can be inherited and subsequently removed without changing the original DNA sequence



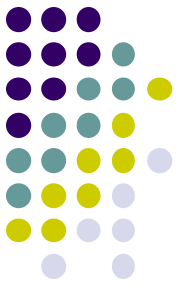
- **CpG sites**: regions of DNA where a cytosine nucleotide occurs next to a guanine nucleotide in the linear sequence of bases along its length.
- C in CpG dinucleotides can be methylated by DNA methyltransferases in many eukaryotic organisms to form 5-methylcytosine.
- ~ 70%-80% of C in **CpG** are methylated.

Frequency of CpGs sites

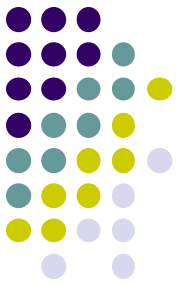


- CpG frequency: 1% in human genome, lower than expected (4X)
- Expected:
 - GC content in human genome: 42%
 - Frequency of G or C: 21%
 - Frequency of CpG: $0.21 \times 0.21 = 4.41\%$
- Why?
 - Transitions (C-T) occur at 10-40 fold higher than other dinucleotides.
- Thus, CpGs are not randomly distributed in the genome.

CpG islands



- **CpG islands:** regions of DNA that have a higher concentration of unmethylated CpG sites.
 - ~ 70%-80% of all CpGs are methylated in mammals, CpGs are methylated in repeats and 3' regions of genes, but not in CpG islands.
 - **Unmethylated CpGs are grouped in clusters called "CpG islands"** that are present in the 5' regulatory regions of many genes.
 - CpG islands are associated with the start of the gene.
 - a CpG island is used to help in the prediction and annotation of genes
- ~ 26,000-45,000 CpG islands, many located in the promoter regions of housekeeping genes and some tissue specific genes.



Methylation, silencing, and cancer

- **Methylation of CpG sites within the promoters of genes can lead to their silencing,**
 - human cancers: the silencing of tumor suppressor genes:

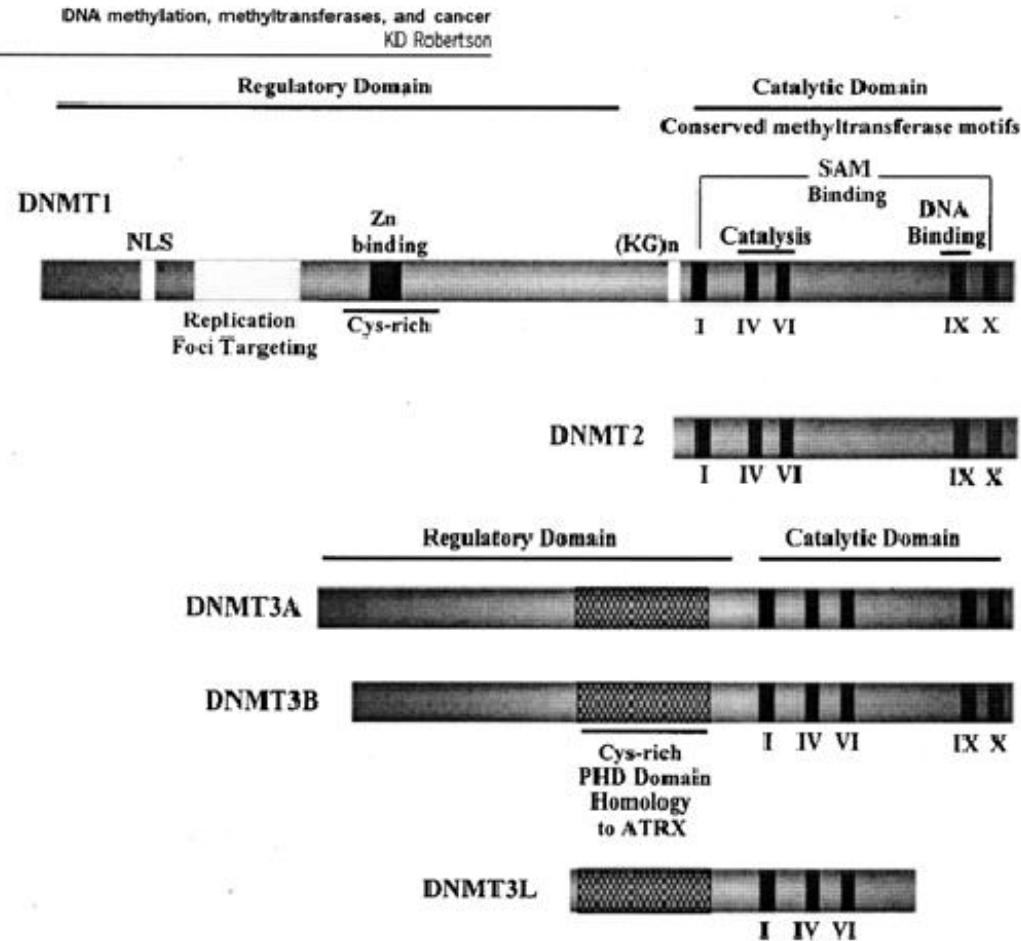
As many as 50% of point mutations in p53 gene in human tumors occur at CpG.
 - In contrast, the hypomethylation (低甲基化) of CpG sites has been associated with the over-expression of oncogenes within cancer cells.

How does methylation occur?



- **De Novo methylation** (重新甲基化): DNA methylation patterns is set up early in development. What signals de novo methylation? Some evidence that repeats may alter chromatin structure and signal de novo methylation.
- **Maintenance Methylation**- DNA methyltransferase (DNMT) is the proposed maintenance methyltransferase that is responsible for copying DNA methylation patterns to the daughter strands during DNA replication.
- **DNA demethylation**: a process of removal of methyl group from nucleotide in DNA.
 - Passive demethylation: takes place in the absence of methylation of newly synthesised DNA strands by DNMT1
 - Active demethylation: Active DNA demethylation occurs via active dismiss of methyl group.

DNA Methylases



-Dnmt 1 thought to be primarily a maintenance methylase. 10-40 fold preference for hemimethylated DNA. Localizes to replication loci.

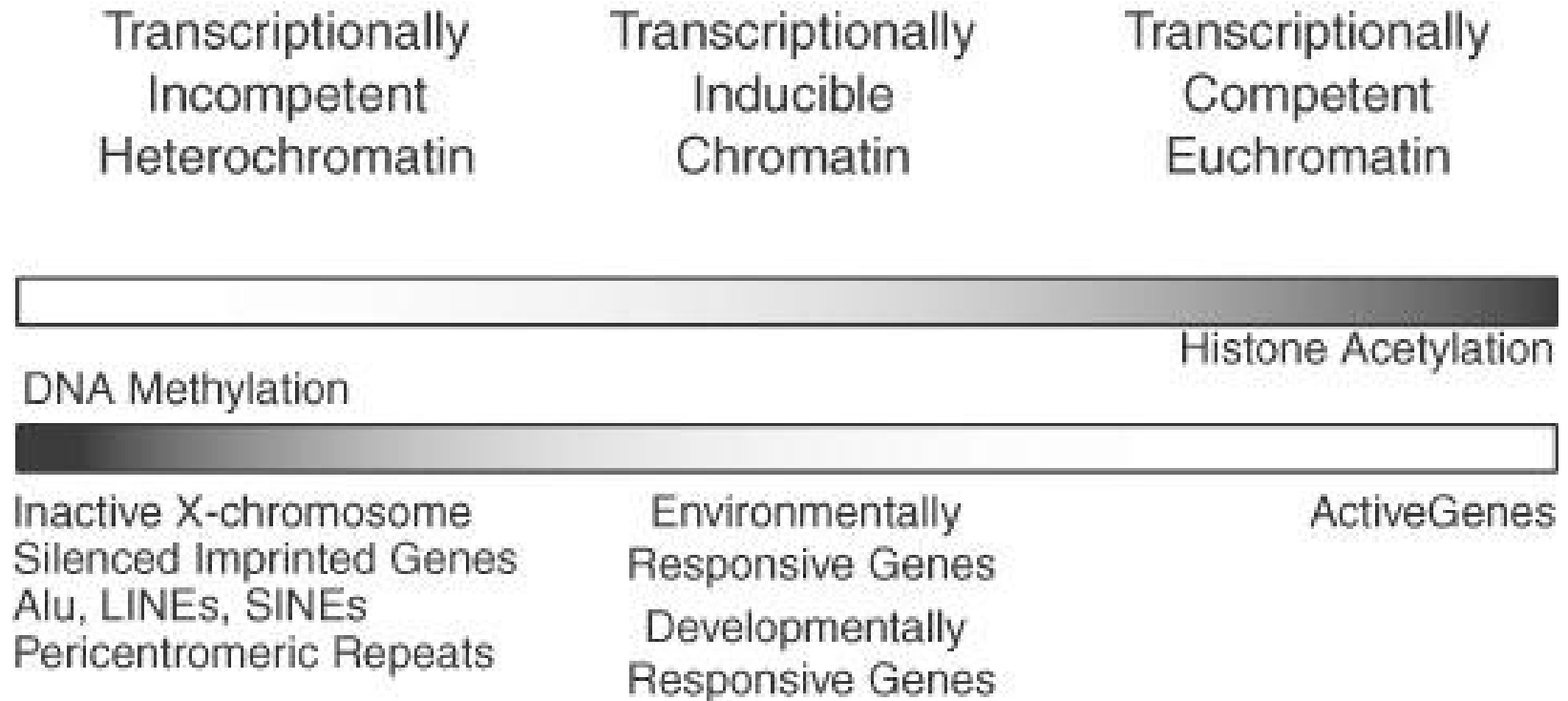
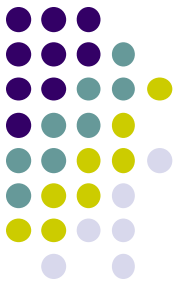
-Dnmt 3A and 3B are thought to be de novo methylases. DNMT3 family contain similar catalytic domain, but different regulatory domains. Contains a PHD (plant homeodomain) found in many chromatin associated proteins.



Dnmts

- The Knock-out of any of the methyltransferase genes is lethal during development (Dnmt 1 or 3B) or shortly thereafter (3A; about 4 weeks of age).
- mutations in Dnmt3B cause ICF syndrome (Immunodeficiency, centromeric instability, facial anomalies).

Interaction between DNA methylation and Histone acetylation



Transcriptionally active DNA- unmethylated DNA with acetylated histones

Repressed (silent) DNA- methylated DNA with deacetylated histones

Prions (感染性蛋白质,朊病毒)

" Prion" is short for "Proteinaceous Infectious particle"

Prions: are infectious forms of proteins. Proteins generally fold into discrete units which perform distinct cellular functions, but some proteins are also capable of forming an infectious conformational state known as a prion.

- Although often viewed in the context of infectious disease,
- Also is defined by their ability to catalytically convert other native state versions of the same protein to an infectious conformational state.
- ~ as epigenetic agents: inducing a phenotypic change without a modification of the genome.

Fungal prions are considered epigenetic because the infectious phenotype caused by the prion can be inherited without modification of the genome.

Transgenerational epigenetic effects

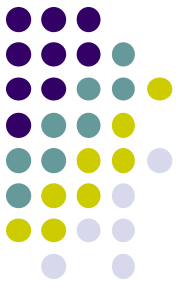
- ~ effects: phenotypes present in successive generations that are not genetically determined.
- ~ inheritance (gametic epigenetic inheritance): a phenotype present in successive generations that is nongenetically determined and results from epigenetic modifications passed via the gametes that escape reprogramming

- Epigenetic modification:

Methylation of CpG + modification of chromatin protein

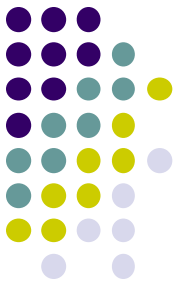
- Erase the epigenetic markers:

Germline and somatic reprogramming



**Transgenerational epigenetic
effects $\neq$ ~ inheritance**

Adaptive parental effect



- **Adaptive maternal effects exist in plants.**

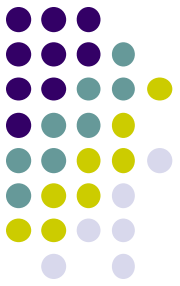
Environmental stimuli acting on the mother, including predation, competition, soil type, temperature, light, and nutrient availability has been found to induce changes to F1 phenotype.

- *Polygonum persicaria* (春蓼)

- offspring of grown under low light allocate proportionately more resources to shoot growth than those of parents with higher light exposures.
- Conversely, offspring of limited nutrient plants allocate proportionately more to root growth than genetically similar individuals with nutrient-rich parents

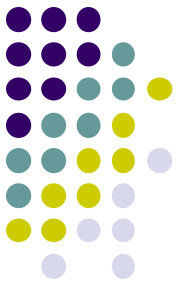
Radishes and caterpillars

(Agrawal, 1999)

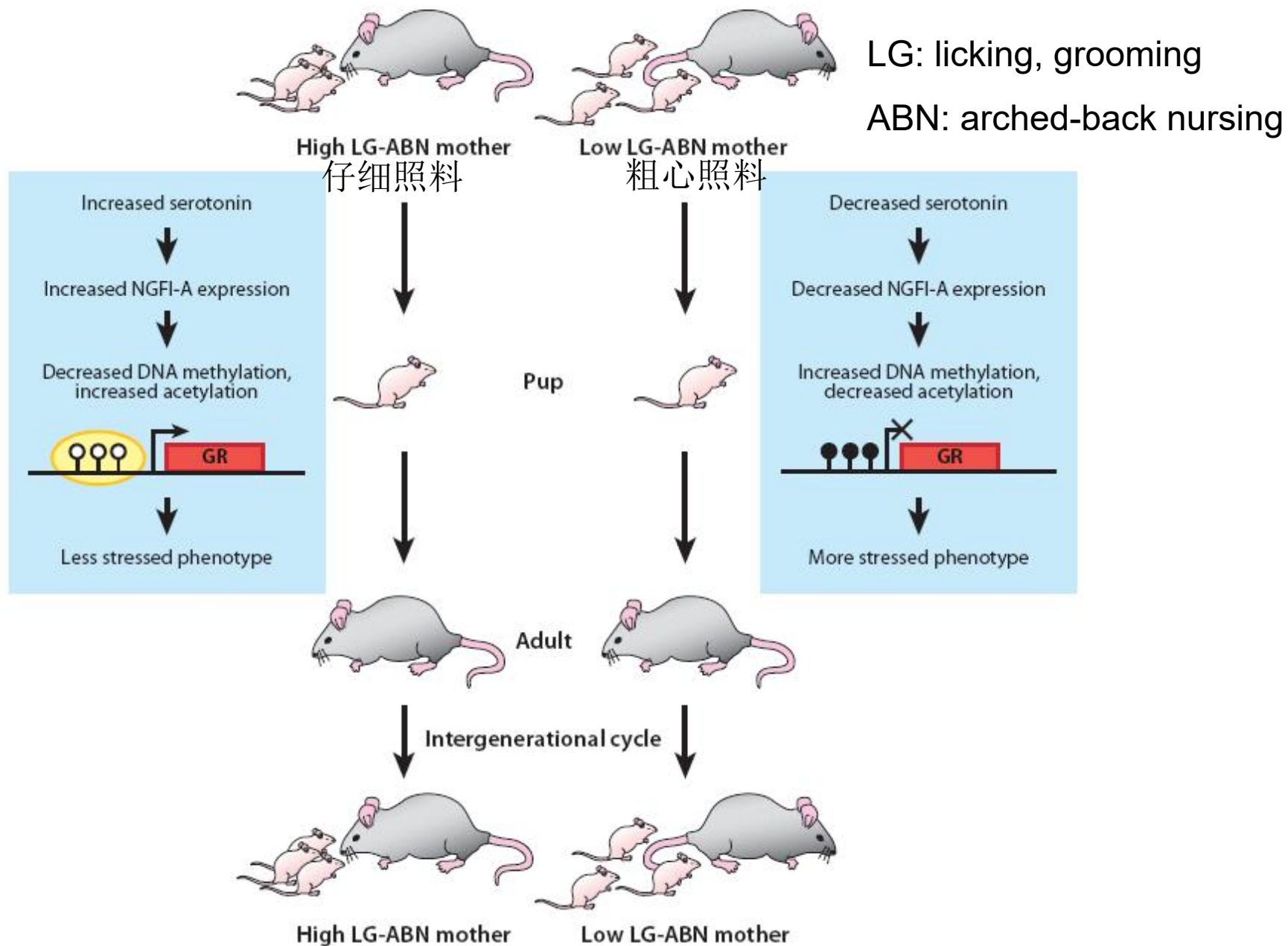


- Radishes produce physical spines and insect-repellent chemicals in response to predation by caterpillars
- Seedlings from parent plants that had been damaged by caterpillars develop phenotypic features (spines) more like their parents than seedlings from unexposed plants.
- These changes to the F1 are associated with reduced predation from caterpillars.

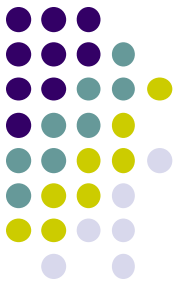
Transgenerational inheritance of mothering style and stress in rat



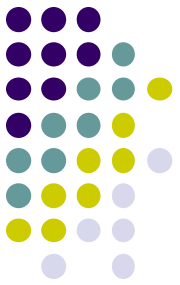
- High LG-ABN mothering leads to increased expression of the transcription factor nerve growth factor inducible protein A (NGFI-A).
- Binding of NGFI-A to the promoter of the glucocorticoid (GR) receptor to stimulates DNA hypomethylation, histone acetylation, and increased expression of GR.
- Higher GR in the hippocampus (海马) are associated with reduced stress levels.
- The epigenetic marks maintain the GR expression state into adulthood and in females will determine the level of LG-ABN mothering, thus perpetuating the phenotype.



Expression of the transcription factor, nerve growth factor inducible protein A (NGFI-A), is controlled by the way how the mother is taking care of the offspring



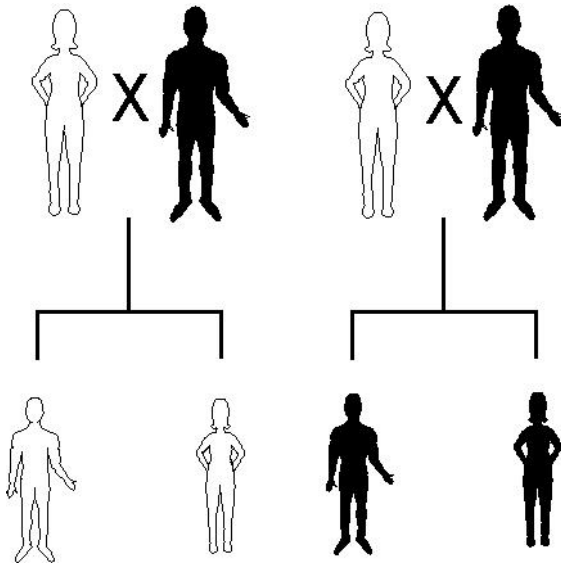
Some interesting examples of *epigenetic inheritance* and *variation*



Too big! Apparently as a result of abnormal *imprinting*, the cloned lamb at left is bigger than the normal lamb at right. Cloned animals often have other health problems as well.



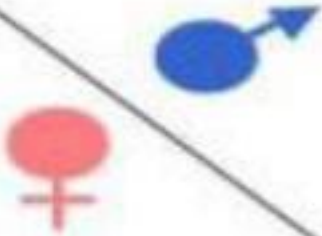
Genomic imprinting can be defined as the gamete-of-origin dependent modification of phenotype._



***Body color
(hypothetical)***

Paternal imprinting: an allele inherited from the father is not expressed in offspring.

Maternal imprinting: an allele inherited from the mother is not expressed in offspring.

	donkey 	horse 
donkey 	donkey 	hinny 
horse 	mule 	horse 

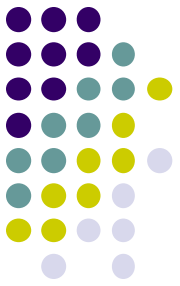
Images credit: <http://www.imaha.org/>

**"parent-of-origin effects" discovered ~3000 years ago
by mule breeders in Asia.**

X-chromosome Inactivation

- **X-inactivation**: a process by which one of the two copies of the X chromosome present in female mammals is inactivated.
 - The inactive X chromosome is silenced by packaging into transcriptionally inactive heterochromatin. X-inactivation occurs so that the female, with two X chromosomes, does not have twice as many X chromosome gene products as the male, which only possess a single copy of the X chromosome
 - The choice of which X chromosome will be inactivated is **random in placental mammals such as mice and humans**, but once an X chromosome is inactivated it will remain inactive throughout the lifetime of the cell.
 - Unlike the random X-inactivation in placental mammals, inactivation in marsupials applies exclusively to the paternally derived X chromosome.

Epigenetic variation among monozygous twins

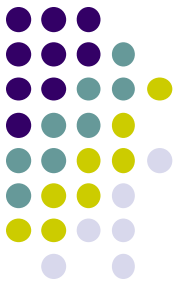


Monozygous twins are considered genetically identical, but significant phenotypic discordance between them exist, which is particularly noticeable for psychiatric diseases.

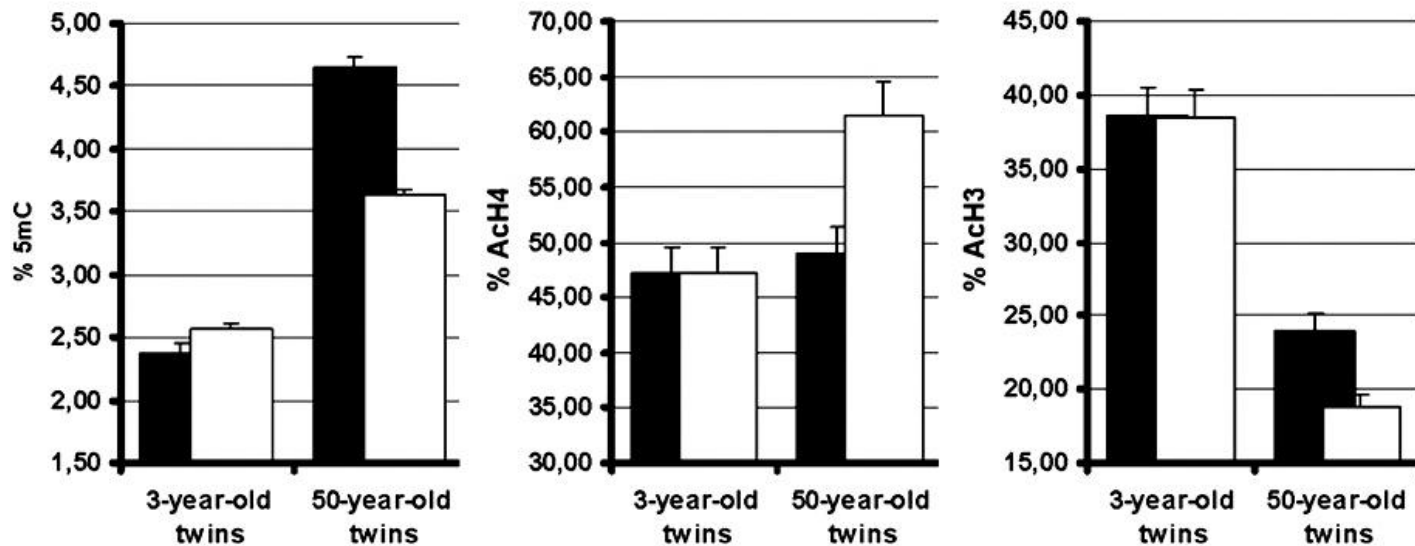


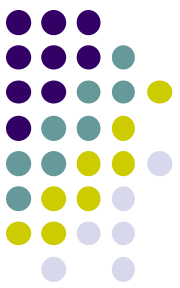
Monozygous Twins with Different Hair Color

Epigenetic variation among monozygous twins



Although MZ twins are epigenetically indistinguishable during early years of life, older MZ twins exhibited remarkable differences in their overall content and genomic distribution of 5-methylcytosine DNA and histone acetylation, affecting their gene expression portrait.





Other cases

你遗传给后代的比你想象的要多
——表观遗传的祖辈“原罪”

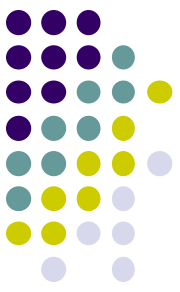
原创2016-06-10丁广进赛先生

受饥荒影响的女性所生后代易发代谢和精神疾病

吃不饱的孕鼠后代个头小易患糖尿病

2014年，来自剑桥大学和哈佛医学院的研究团队发表在《科学》上的论文通过使用小鼠模型，模拟了妊娠期营养不良对后代的影响（给怀孕**12.5**天到**18.5**天的小鼠喂食比平时少**50%**热量的食物），发现该种情况下孕育的后代（**F1**代）个体较小、且更易患糖尿病，令人更为惊讶的是这些雄性小鼠（**F1**代，在没有饮食限制的前提下）的后代（**F2**代）也出现了相同的症状。

Science, 345(6198), 1255903.



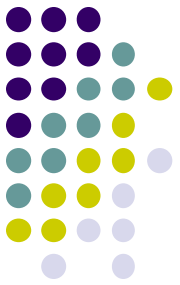
吃太好所生后代出现代谢紊乱

在2014年12月4日发表在《细胞》（**Cell**）杂志上的封面文章中，研究人员在雄性果蝇交配的前两天内增加其饮食的糖含量，然后使之与正常饮食的雌果蝇交配。雌果蝇产下的卵经过正常发育至成年后，与正常营养条件下繁殖的后代相比，其体重偏大且甘油三酯水平显著升高。 **Cell**, 159(6), 1352-1364.

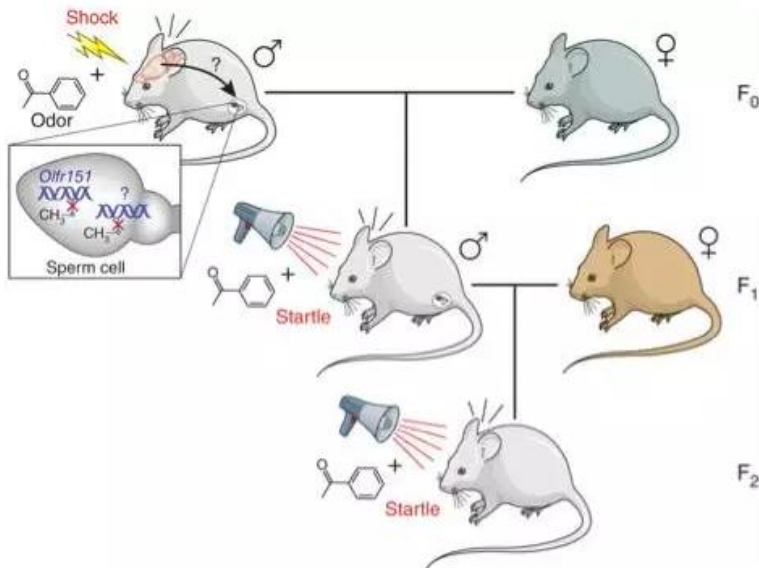
2015年年底，来自中科院段恩奎、翟琦巍、周琪等课题组合作发表在《**Science**》上的研究，运用高脂饲喂诱导的父代肥胖小鼠模型，发现一类成熟精子中高度富集的小RNA(**tsRNAs**)可作为一种表观遗传信息的载体，将高脂诱导的父代代谢紊乱表型遗传给子代。

Science. 2016, 351(6271):397-400.

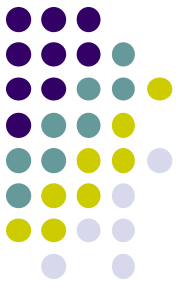
父辈的恐惧情绪记忆可遗传至子代



2013年，发表在《Nature Neuroscience》上的一篇论文激起表观遗传学领域的专家学者热烈的讨论。在这篇文章中，来自美国埃默里大学医学院等研究机构人员通过使用电击等手段使实验鼠对某种气味（苯乙酮）产生恐惧，然后对他们进行正常饲养。结果发现实验组小鼠的后代都会对这种气味非常敏感，并表现出恐惧感，即便是通过人工授精获得的后代（第三代）也不例外。



Nature neuroscience, 17(1), 89-96.



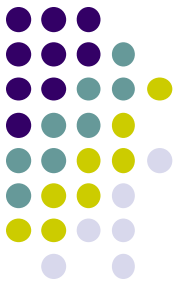
“组蛋白去甲基化酶**KDM6B**调控了红耳龟的温度依赖型性别决定”



红耳龟在较低的孵化温度(26°C)产生 100%的雄性子代, 较高的温度(32°C)则产生全雌子代, 是 TSD 机制研究的一个理想动物模型。

在性腺分化前, Kdm6b 能快速响应温度变化。**KDM6B**通过消除启动子区**H3K27me3**标记, 直接促进红耳龟雄性性别决定基因*Dmrt1*的表达。

浙江万里学院葛楚天等 **Science. 2018.5**



Science重要发现：母爱竟会改变孩子DNA

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REPORT



Early life experience drives structural variation of neural genomes in mice

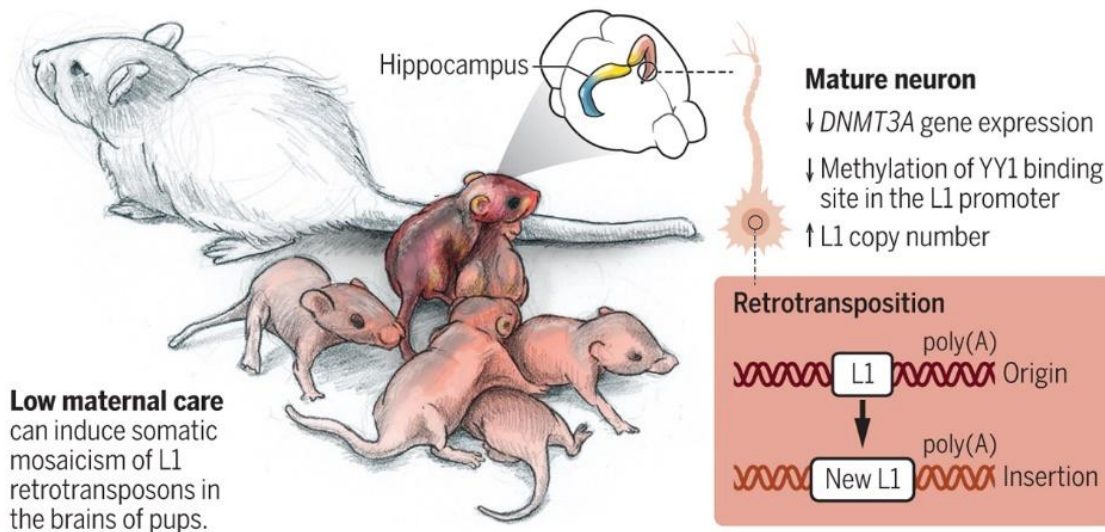
Tracy A. Bedrosian^{*}, Carolina Quayle, Nicole Novaresi, Fred. H. Gage^{*}

⁺ See all authors and affiliations

Science 23 Mar 2018:
Vol. 359, Issue 6382, pp. 1395-1399
DOI: 10.1126/science.aah3378

Maternal care alters genomic structure

Early life experiences such as maternal care affect DNA sequence in neurons of the hippocampus via L1 retrotransposition. The accumulation of L1 retrotransposons in the hippocampus of rodent pups reared with low maternal care might contribute to higher anxiety-like behavior in adulthood.

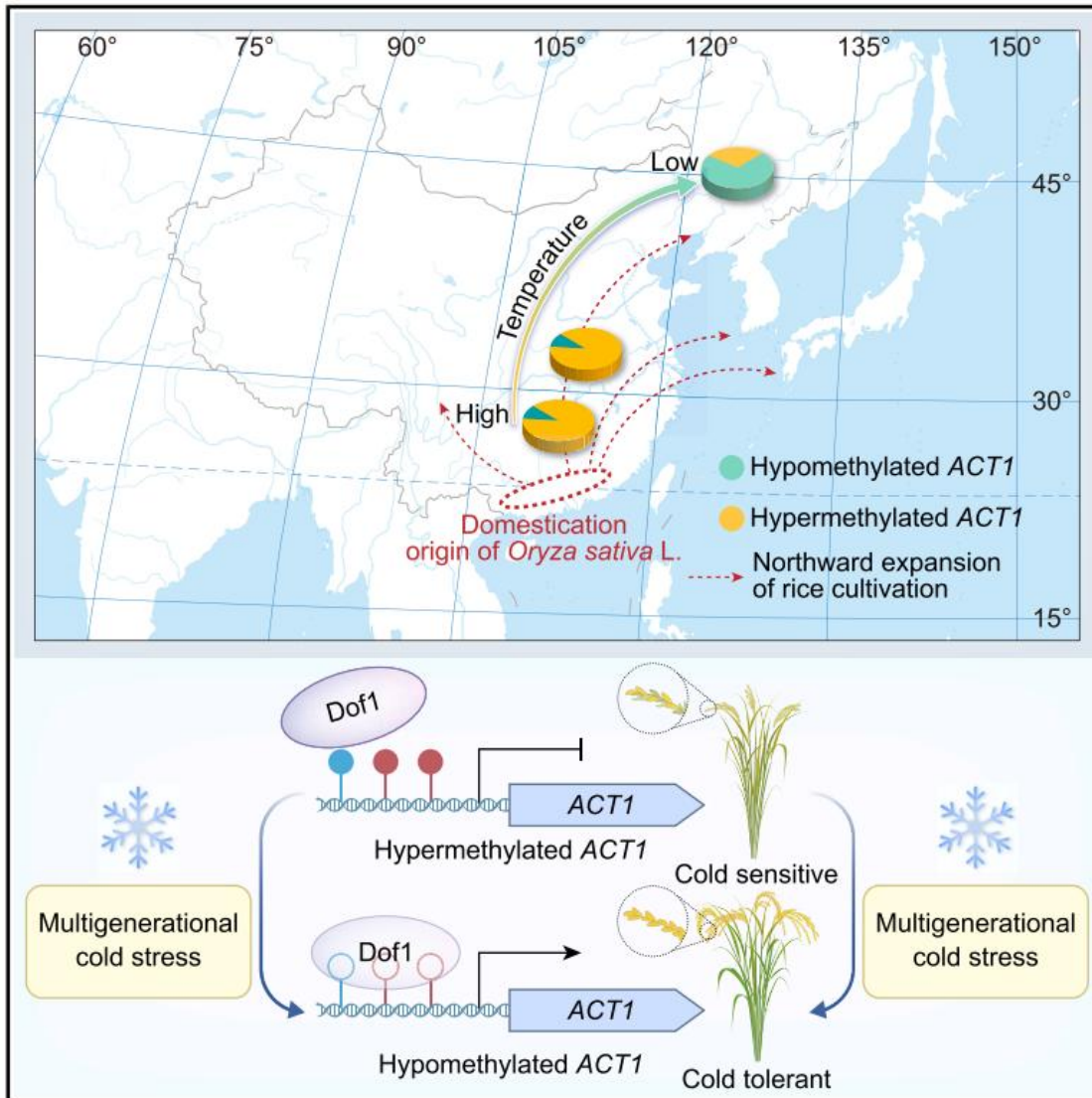


母亲的关怀（maternal care）与海马体的L1拷贝数量之间存在相关性：拥有细心母亲的小鼠携带的L1拷贝数较少，而拥有疏忽大意母亲的小鼠拥有更多的L1拷贝。“生母疏忽大意但养母细心的小鼠”比“生母细心但养母疏忽大意的小鼠”拥有更少L1拷贝。

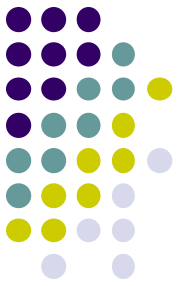


越来越多的证据表明，在不改变遗传物质**DNA**序列的情况下，上一辈可以将不良营养习惯和环境压力产生的某些获得性性状遗传给下一代，这种传递通常是通过表观遗传的方式进行的。值得注意的是，目前在哺乳动物中获得的实验数据表明，获得性性状的遗传很难像经典遗传学中的基因突变持续遗传下去，实验数据表明这种遗传很难超过三代，但是不可否认这种遗传方式具有重要意义。人们应该从这些科学研究中，认识到自身的生活饮食习惯、所处环境、心态情绪等对下一代可能造成的影响。

Inheritance of acquired adaptive cold tolerance in rice through DNA methylation



Multigenerational cold stress induces heritable ACT1 promoter hypomethylation, enabling rice to acquire cold tolerance for high-latitude adaptation.



母亲孕期铁元素缺乏竟能让儿子变女儿！

[nature](#) > [articles](#) > [article](#)

Article | Published: 04 June 2025

Maternal iron deficiency causes male-to-female sex reversal in mouse embryos

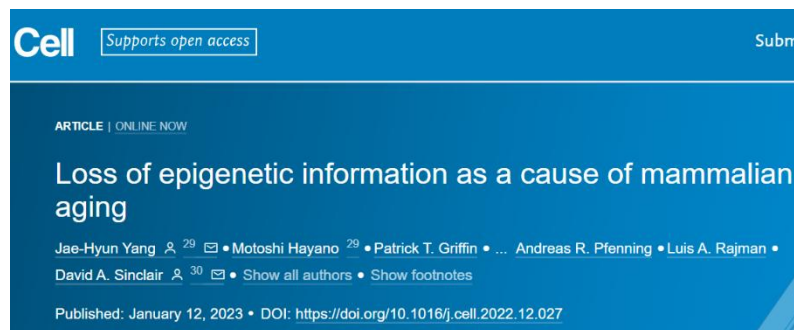
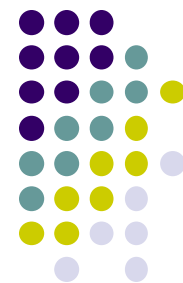
[Naoki Okashita](#), [Ryo Maeda](#), [Shunsuke Kuroki](#), [Kyona Sasaki](#), [Yoko Uno](#), [Peter Koopman](#) & [Makoto Tachibana](#) 

[Nature](#) (2025) | [Cite this article](#)

DOI: 10.1038/s41586-025-09063-2

在孕期缺铁的小鼠中，部分本应发育为雄性的胚胎，竟发育出卵巢结构，最终表现出雌性表型。

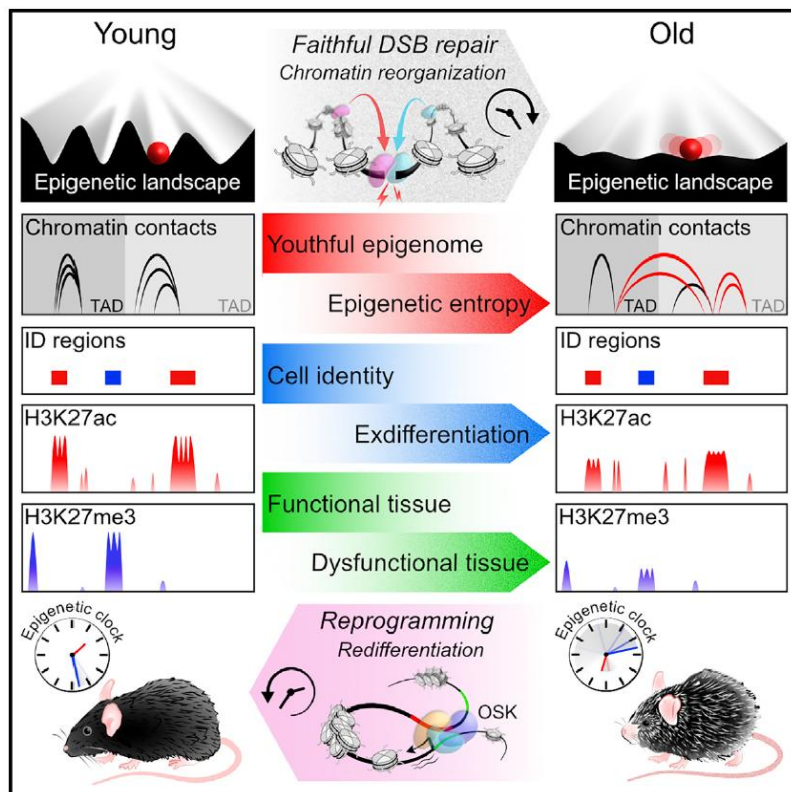
表观遗传信息的丢失是哺乳动物衰老的原因之一

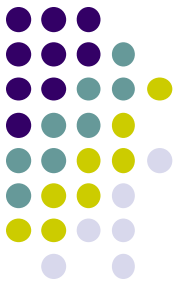


哈佛医学院 David Sinclair

真核生物的衰老是由于转录网络和表观遗传信息随着时间的推移而丢失，由一种保守机制驱动，该机制进化为共同调节对细胞损伤的反应，如**DSB**或挤压损伤。

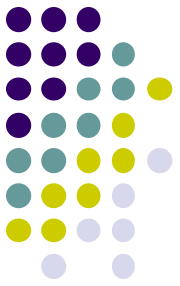
DNA变化不是衰老的唯一原因。相反，染色质的化学和结构变化在不改变遗传密码本身的情况下加速衰老。





你是你，也可以是更好的你！

Thank you for your support ,
wish you all GREAT SUCCESS !



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